



SCHOOL OF FUTURE TECH

Case Study Report

On

Harbor Lane Legal Group

Network Foundations, Topology & IP
Addressing Design

Computer Networks

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CASE STUDY REPORT

HARBOR LANE LEGAL GROUP

Network Foundations, Topology & IP Addressing Design

Computer Networking

Submitted By:

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Tools Used: Cisco Packet Tracer

Project Type: Enterprise Network Design and Implementation

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1. ABSTRACT

This project focuses on designing and implementing an enterprise network infrastructure for Harbor Lane Legal Group. The organization requires a reliable and scalable network connecting its Head Office with two branch offices, Riverside and Uptown.

The proposed network uses VLAN segmentation to separate the major departments at the Head Office. VLSM is used to allocate IPv4 addresses according to the host requirements of each network segment. Router-on-a-Stick is implemented to provide communication between VLANs, while trunk links carry multiple VLANs between the core and departmental switches.

The network also includes serial WAN leased-line connectivity between the Head Office and branch offices. A dedicated Voice VLAN is implemented for VoIP communication, with DHCP providing automatic IP addressing to the IP phone.

The complete network was designed, configured, and tested using Cisco Packet Tracer. Connectivity was verified using gateway ping tests, inter-VLAN communication tests, WAN connectivity tests, DHCP verification, interface status verification, and routing-table inspection.

2. INTRODUCTION

Modern organizations require secure, scalable, and reliable computer networks to support daily operations. A legal organization such as Harbor Lane Legal Group handles sensitive information and therefore requires proper network segmentation, controlled communication, reliable connectivity, and an organized IP addressing structure.

The organization consists of a Head Office and two satellite offices:

- Head Office
- Riverside Office
- Uptown Office

The Head Office contains four major departments:

1. Litigation Department
2. Corporate Law Department
3. IT & Records Department

4. Reception/Admin Department

Each major Head Office department is assigned a separate VLAN. This logical separation reduces unnecessary broadcast traffic and provides a structured foundation for network management.

The network also supports printers, employee systems, a VoIP phone, and WAN connectivity to the two branch offices. The overall implementation follows an extended-star architecture at the Head Office with serial point-to-point WAN connections to the branches.

3. PROBLEM STATEMENT

Harbor Lane Legal Group is expanding its operations and requires a properly planned network infrastructure before deploying networking equipment.

The network design must provide:

- Separate networks for different departments
- Efficient IP address allocation
- Communication between departments when required
- Connectivity between Head Office and branch offices
- Support for printers and VoIP devices
- Reliable network connectivity
- Easy troubleshooting and management
- Scope for future expansion

The network should minimize unnecessary broadcast traffic while maintaining communication between required network segments.

4. PROJECT OBJECTIVES

The main objectives of this project are:

4.1 Network Segmentation

Separate the major departments using VLANs to logically isolate their network traffic.

4.2 Efficient IP Address Management

Use VLSM to allocate subnet sizes according to the number of required hosts while maintaining spare capacity.

4.3 Inter-VLAN Communication

Implement Router-on-a-Stick to allow communication between different VLANs through the Head Office router.

4.4 WAN Connectivity

Establish point-to-point serial WAN connections between the Head Office and Riverside and Uptown offices.

4.5 Voice Communication

Create a dedicated Voice VLAN for the IP phone and provide automatic addressing using DHCP.

4.6 Network Verification

Verify the implementation using:

- Ping tests
- DHCP verification
- Interface status
- Routing-table verification
- Inter-VLAN communication
- WAN connectivity tests

These objectives correspond to the implemented network architecture and testing evidence in the project.

5. NETWORK REQUIREMENTS ANALYSIS

The network requirements provided for the case study are:

Network Segment	Required Hosts	Purpose
HO – Litigation	45	Attorney laptops and e-discovery workstations

HO – Corporate Law	33	Corporate legal employees
HO – IT & Records	9	IT resources and records management
HO – Reception/Admin	4	Reception and administrative devices
HO – Riverside WAN	2	Point-to-point WAN connection
HO – Uptown WAN	2	Point-to-point WAN connection

A dedicated Voice VLAN was additionally implemented to support the VoIP phone located in Reception/Admin.

6. PHYSICAL NETWORK TOPOLOGY DESIGN

6.1 Topology Recommendation

The implemented Head Office network follows an **Extended Star Topology**.

A central core switch acts as the backbone of the Head Office. Departmental access switches connect to the core switch, while end devices connect to their respective departmental switches.

The Head Office router performs:

- Inter-VLAN routing
- VLAN gateway functions
- WAN connectivity
- Voice VLAN gateway functions

The branch offices use a smaller star-based LAN arrangement connected through their respective routers.

This architecture provides:

- Easy addition of new devices
- Centralized network management
- Easier troubleshooting
- Departmental traffic separation
- Predictable connectivity paths

- Reasonable fault isolation

A failure of an individual access link normally affects the connected device or segment, while the core switch and Head Office router remain critical infrastructure components.

6.2 Base Network Device Placement

The initial Packet Tracer workspace was created by placing the routers, switches, and end devices required for the enterprise network.

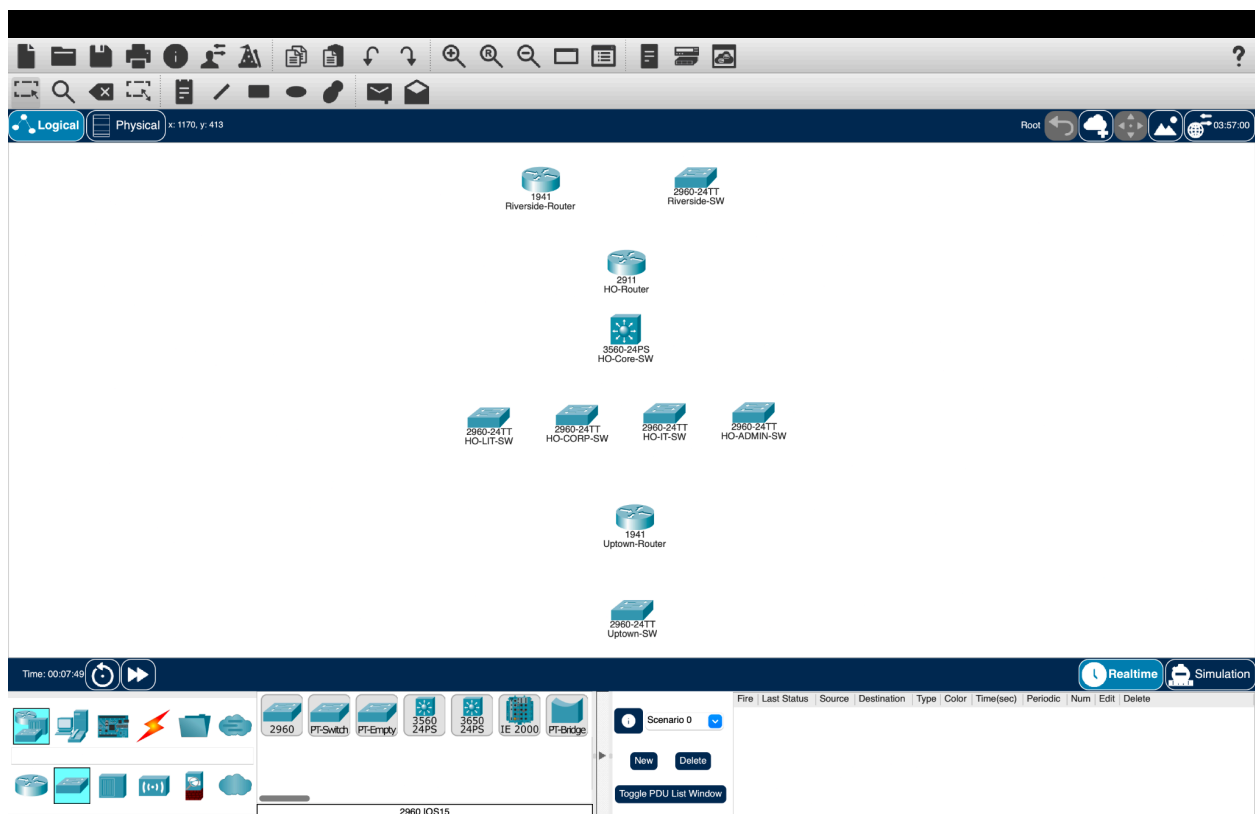


Figure 1: Initial Packet Tracer topology showing the placement of routers, switches, and network devices.

6.3 End Device Placement

The departmental end devices were placed according to the network requirements. These include employee laptops, desktop systems, printers, and the IP phone.

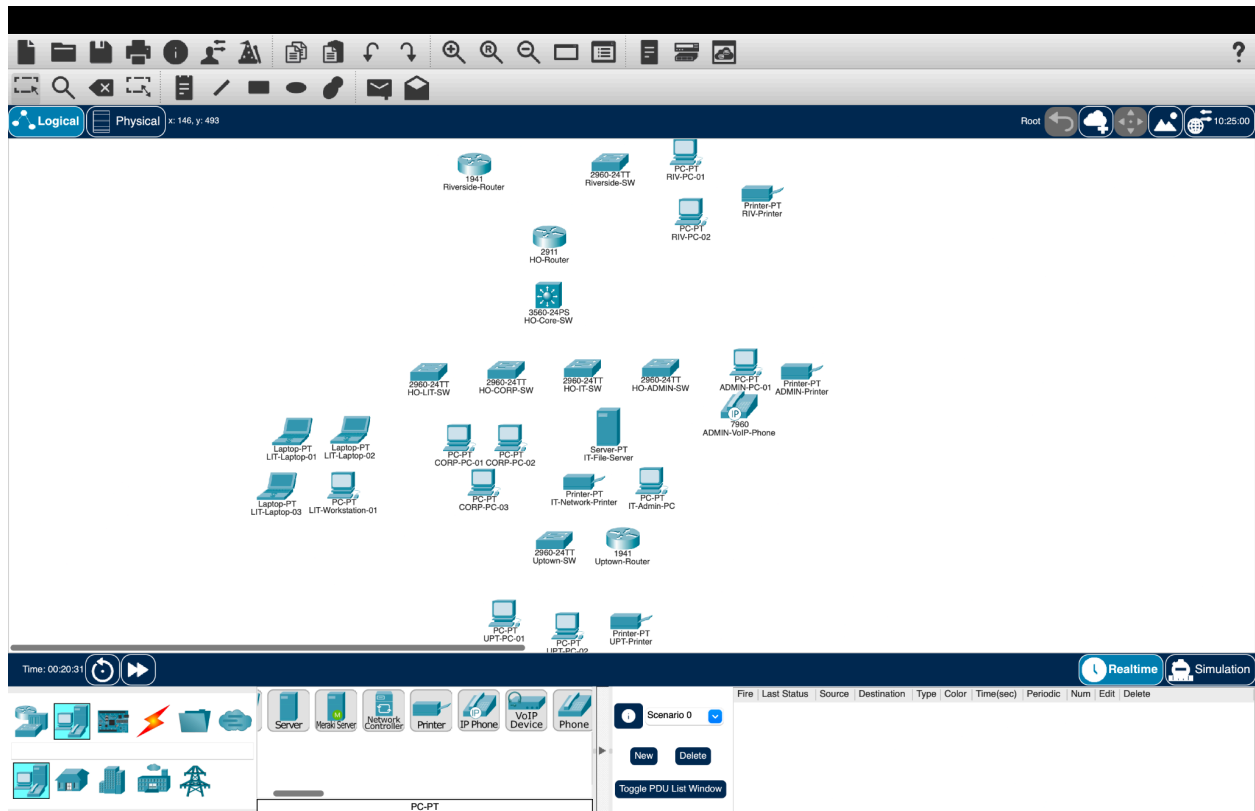


Figure 2: Placement of end devices across the different departments.

7. HEAD OFFICE BACKBONE AND DEPARTMENTAL NETWORKS

7.1 Head Office Backbone

The Head Office uses a centralized backbone design.

The core switch connects:

- Litigation Switch
- Corporate Law Switch
- IT & Records Switch
- Reception/Admin Switch
- Head Office Router

Trunk connectivity is used to carry VLAN traffic between the relevant network devices.

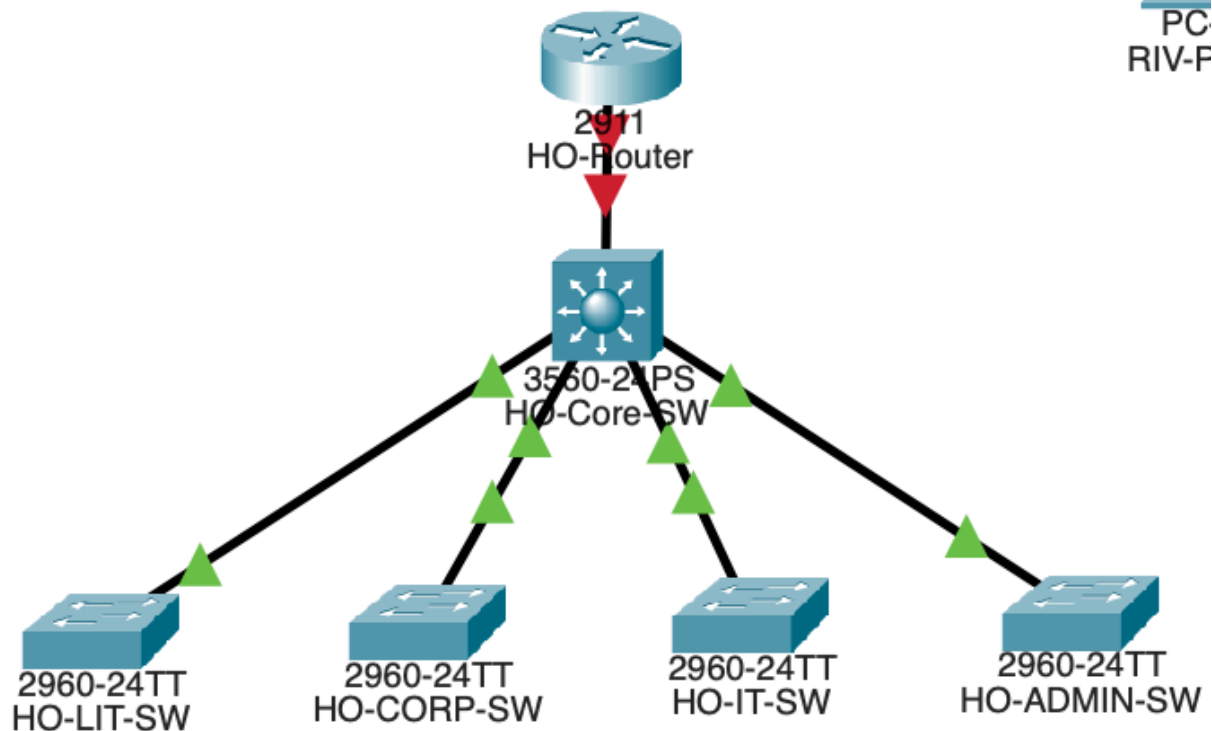


Figure 3: Head Office backbone connections between the core switch, router, and departmental switches.

7.2 Litigation Department

The Litigation Department contains attorney and workstation devices connected through a dedicated access switch.

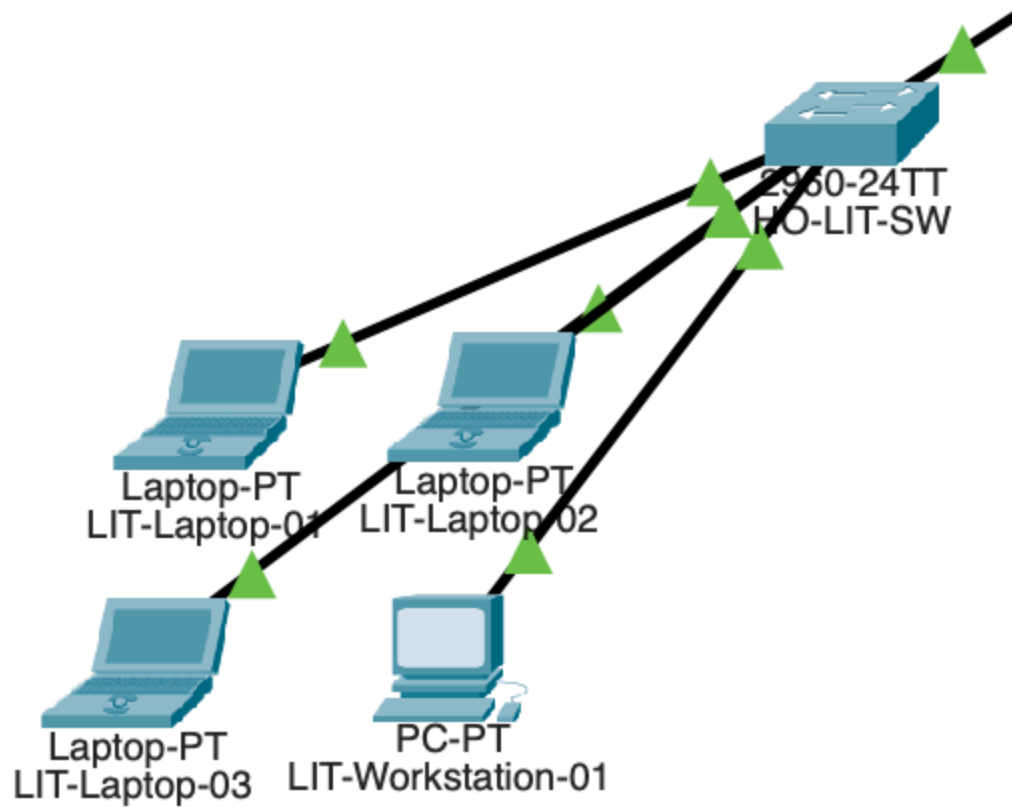


Figure 4: HO-LIT-SW and connected Litigation department devices.

7.3 Corporate Law Department

The Corporate Law Department is connected through a dedicated departmental switch, providing a separate logical network segment.

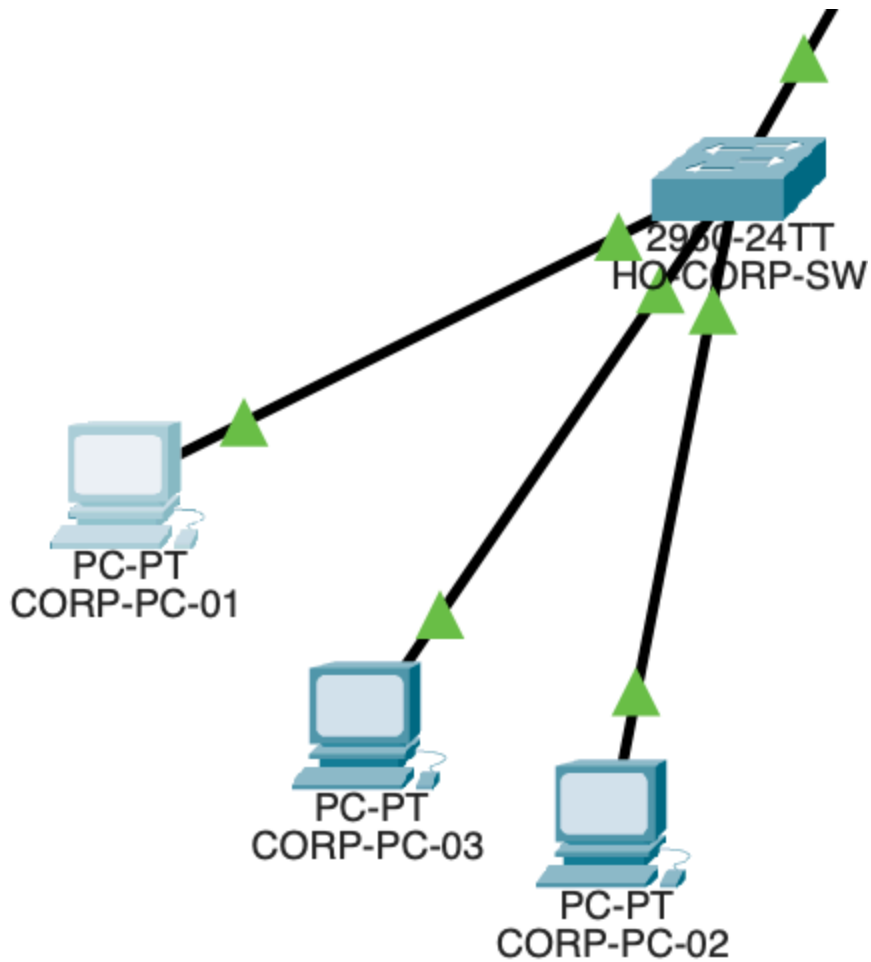


Figure 5: HO-CORP-SW and Corporate Law department connections.

7.4 IT & Records Department

The IT & Records Department contains the systems used for technical resources and records management.

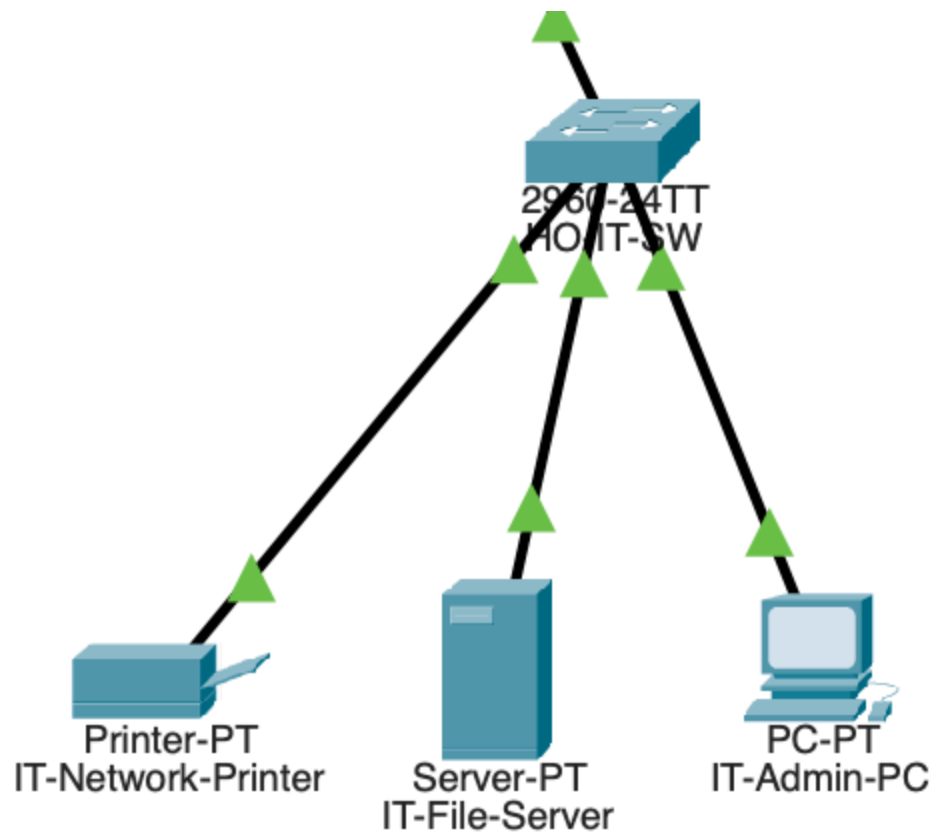


Figure 6: HO-IT-SW and IT & Records department connectivity.

7.5 Reception/Admin Department

The Reception/Admin Department supports administrative devices, a printer, and the VoIP phone.

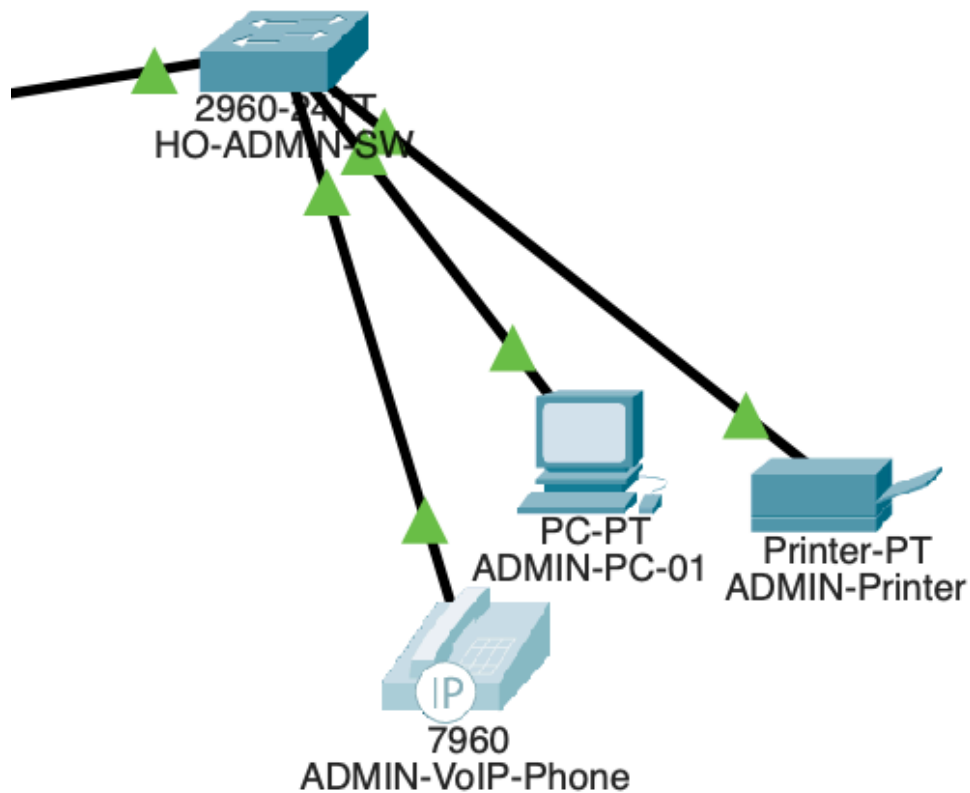


Figure 7: HO-ADMIN-SW and Reception/Admin device connections.

8. BRANCH OFFICE NETWORK DESIGN

8.1 Riverside Office

The Riverside branch follows a small star-based LAN architecture consisting of a router, switch, and connected end devices.

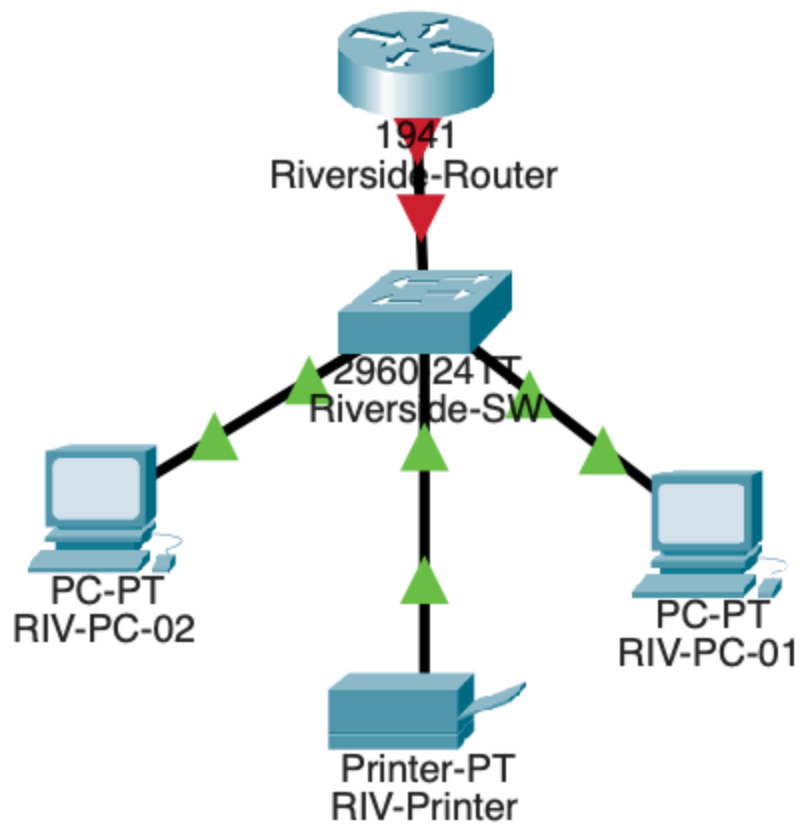


Figure 8: Riverside branch LAN connections.

8.2 Uptown Office

The Uptown branch follows a similar small LAN architecture consisting of a router, switch, and connected end devices.

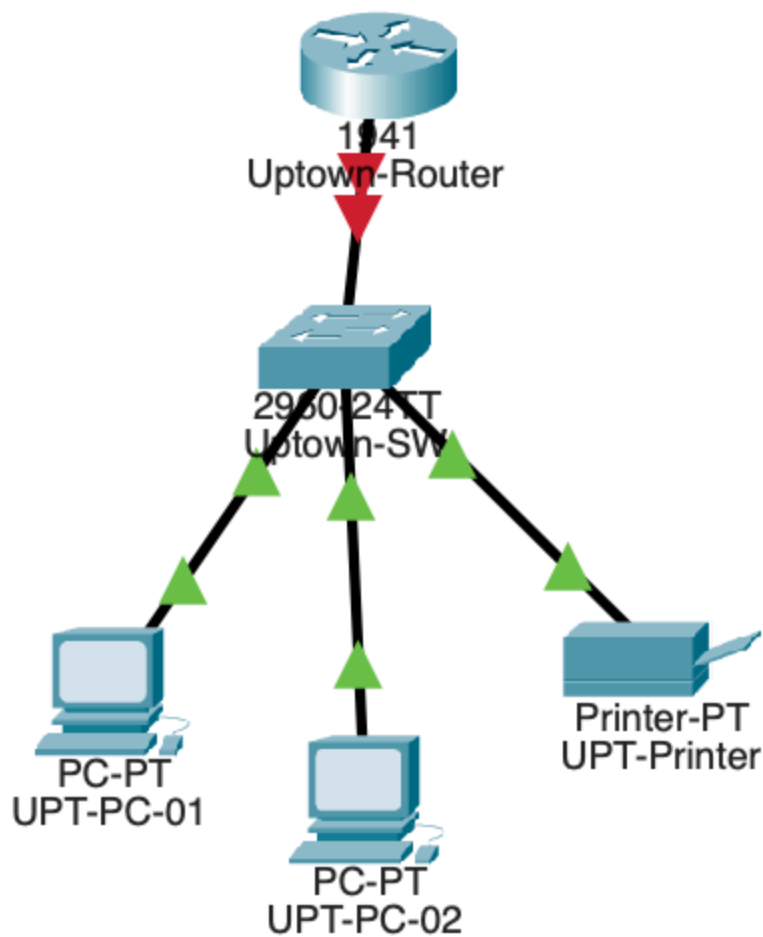


Figure 9: Uptown branch LAN connections.

9. WAN CONNECTIVITY

9.1 Serial Module Installation

The Head Office router was prepared for serial WAN connectivity by installing a suitable serial interface module.

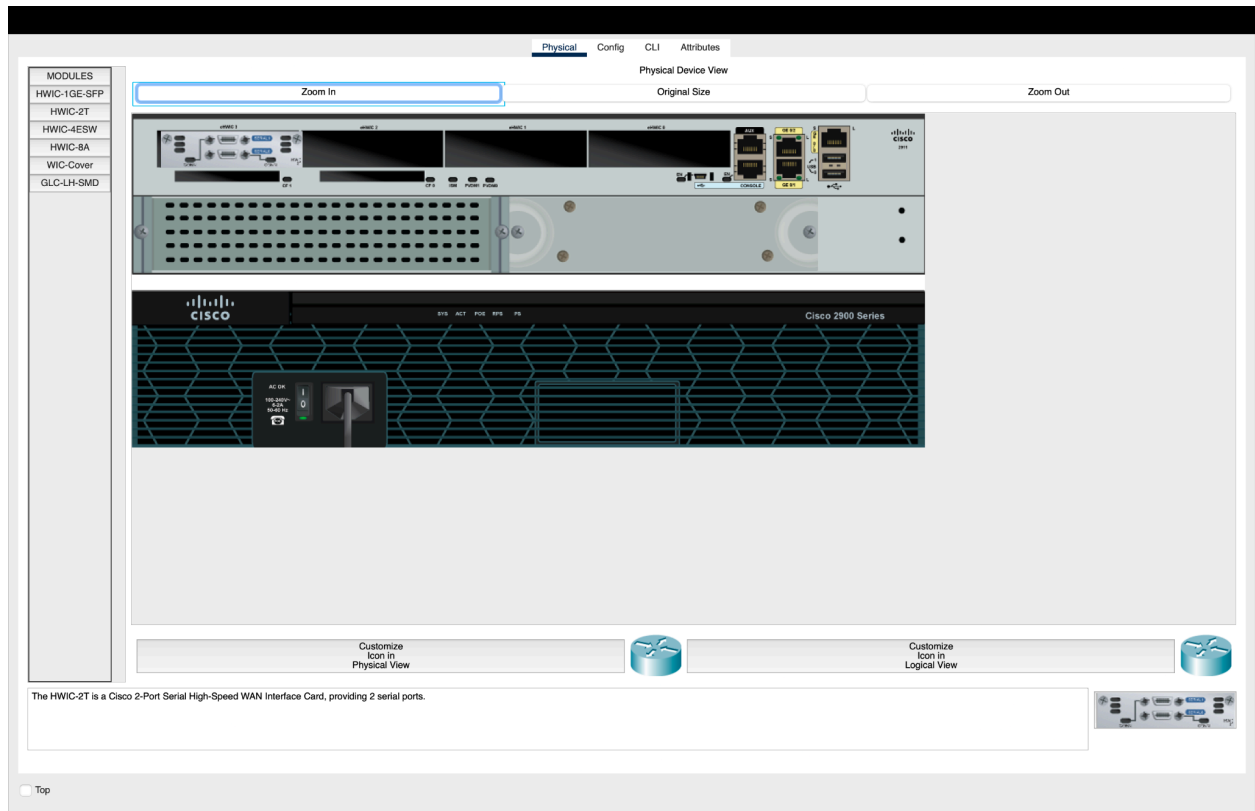


Figure 10: Installation of the HWIC-2T serial interface module on the Head Office router.

9.2 WAN Physical Connectivity

The Head Office is connected to Riverside and Uptown using serial point-to-point leased-line connections.

The WAN routers are:

- HO-Router
- Riverside-Router
- Uptown-Router

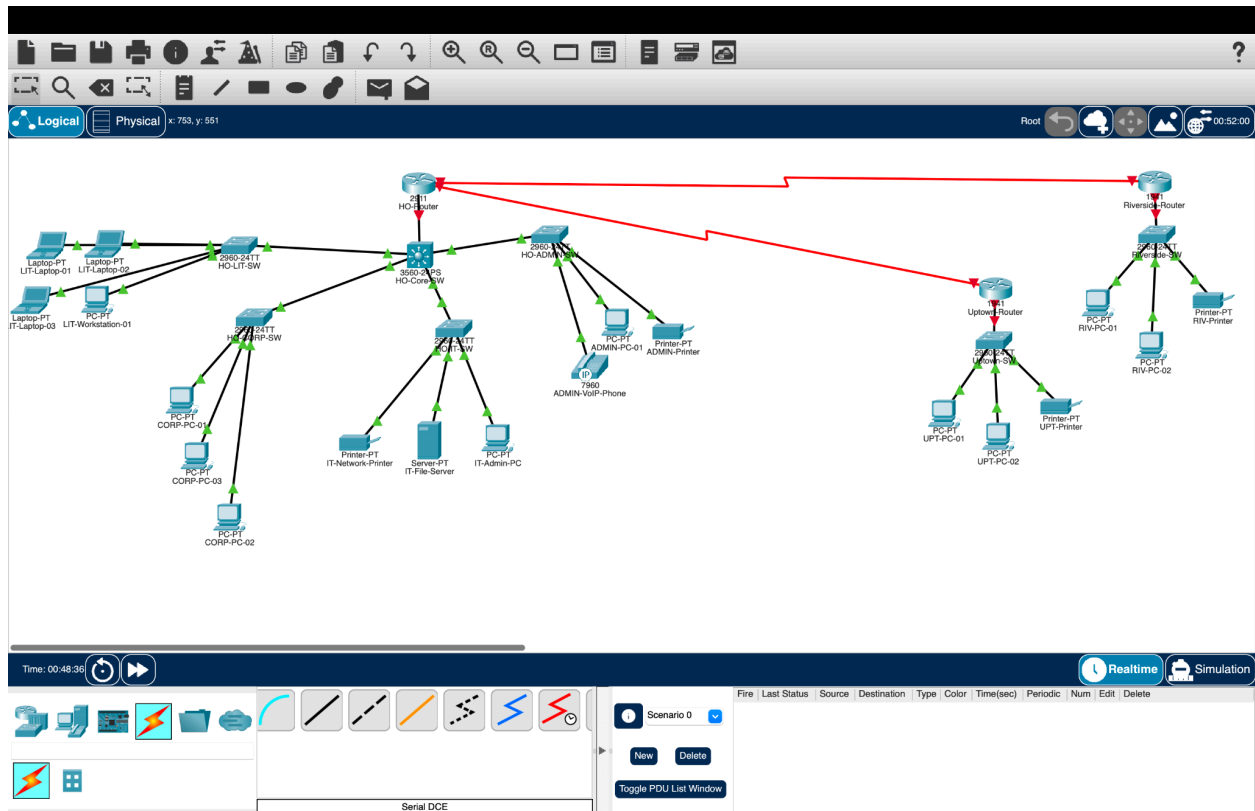


Figure 11: Physical serial WAN connections between the three office routers.

9.3 Head Office WAN Interface Configuration

The Head Office router uses two serial interfaces:

Interface	IP Address	Purpose
Serial0/3/0	192.168.43.1/30	HO → Riverside
Serial0/3/1	192.168.43.5/30	HO → Uptown

Physical Config CLI Attributes

IOS Command Line Interface

```
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
HO-Router#enable
HO-Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
HO-Router(config)#
HO-Router(config)#interface serial0/3/0
HO-Router(config-if)#ip address 192.168.43.1 255.255.255.252
HO-Router(config-if)#clock rate 64000
HO-Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
HO-Router(config-if)#exit
HO-Router(config)#
HO-Router(config)#interface serial0/3/1
HO-Router(config-if)#ip address 192.168.43.5 255.255.255.252
HO-Router(config-if)#clock rate 64000
HO-Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/1, changed state to down
HO-Router(config-if)#exit
HO-Router(config)#
HO-Router(config)#end
HO-Router#write
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
HO-Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0    unassigned      YES unset  administratively down  down
GigabitEthernet0/1    unassigned      YES unset  administratively down  down
GigabitEthernet0/2    unassigned      YES unset  administratively down  down
Serial0/3/0          192.168.43.1    YES manual  down          down
Serial0/3/1          192.168.43.5    YES manual  down          down
Vlan1              unassigned      YES unset  administratively down  down
HO-Router#
```

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Figure 12: HO-Router serial WAN interface configuration.

9.4 Riverside WAN Interface

The Riverside router uses:

Serial0/1/0 — 192.168.43.2/30

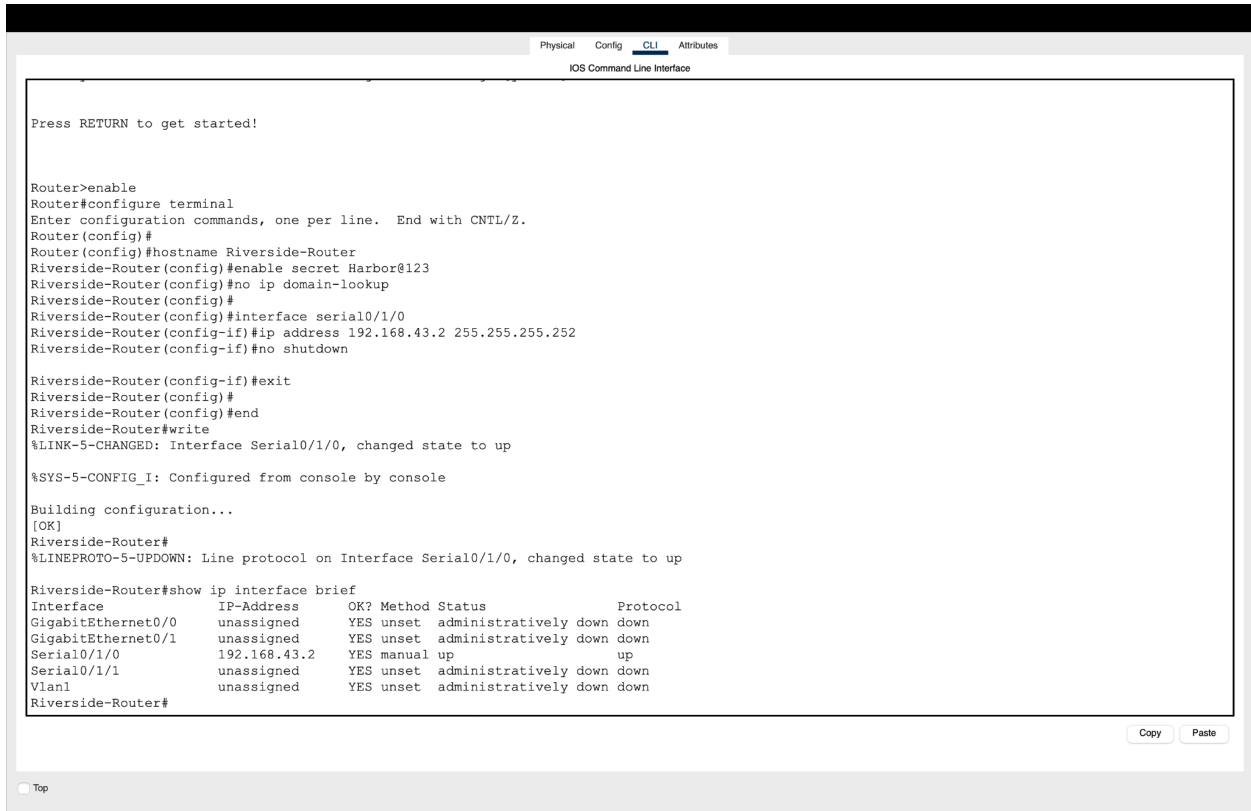


Figure 13: Riverside-Router WAN interface configuration.

9.5 Uptown WAN Interface

The Uptown router uses:

Serial0/1/0 — 192.168.43.6/30

```

Uptown-Router#show ip interface brief
Interface      IP-Address      OK? Method Status        Protocol
GigabitEthernet0/0    unassigned      YES unset  administratively down  down
GigabitEthernet0/1    unassigned      YES unset  administratively down  down
Serial0/1/0          192.168.43.6    YES manual  up              up
Serial0/1/1          unassigned      YES unset  administratively down  down
Vlan1              unassigned      YES unset  administratively down  down
Uptown-Router#

```

Figure 14: Uptown-Router WAN interface configuration.

9.6 WAN Connectivity Verification

The WAN links were verified using ICMP ping tests.

The Head Office router successfully reached:

- Riverside: 192.168.43.2
- Uptown: 192.168.43.6

```
HO-Router#ping 192.168.43.2
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 192.168.43.2, timeout is 2 seconds:
```

```
!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 3/4/12 ms
```

```
HO-Router#ping 192.168.43.6
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 192.168.43.6, timeout is 2 seconds:
```

```
!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/6/19 ms
```

Figure 15: WAN connectivity verification between the Head Office and branch routers.

10. VLAN IMPLEMENTATION

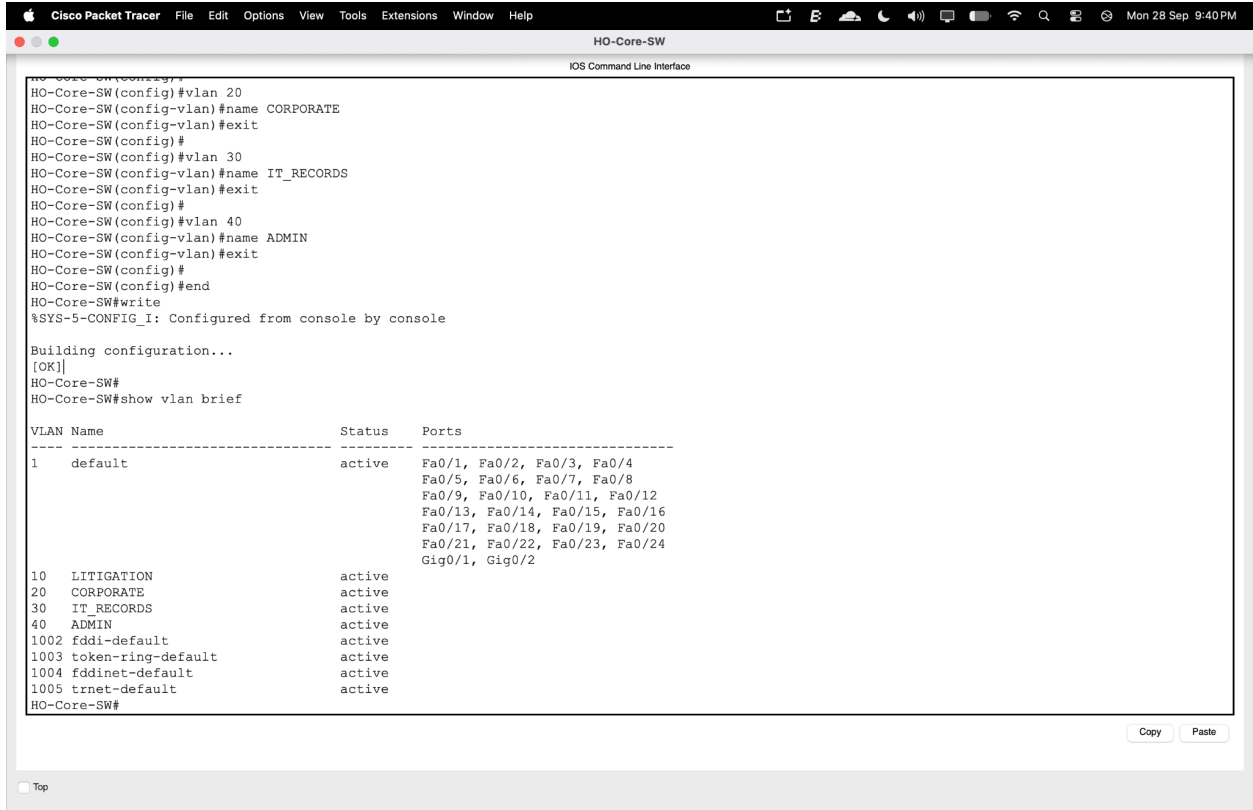
VLAN segmentation was implemented at the Head Office.

VLAN ID	VLAN Name	Purpose
10	LITIGATION	Litigation Department
20	CORPORATE	Corporate Law Department
30	IT	IT & Records
40	ADMIN	Reception/Admin
50	VOICE	IP Phone

VLANs allow the organization to separate traffic logically while using the same physical switching infrastructure.

10.1 Core Switch VLAN Creation

The required VLANs were created on the Head Office core switching infrastructure.



```
HO-Core-SW(config)#vlan 20
HO-Core-SW(config-vlan)#name CORPORATE
HO-Core-SW(config-vlan)#exit
HO-Core-SW(config)#
HO-Core-SW(config)#vlan 30
HO-Core-SW(config-vlan)#name IT_RECORDS
HO-Core-SW(config-vlan)#exit
HO-Core-SW(config)#
HO-Core-SW(config)#vlan 40
HO-Core-SW(config-vlan)#name ADMIN
HO-Core-SW(config-vlan)#exit
HO-Core-SW(config)#
HO-Core-SW(config)#end
HO-Core-SW#write
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
HO-Core-SW#
HO-Core-SW#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	LITIGATION	active	
20	CORPORATE	active	
30	IT_RECORDS	active	
40	ADMIN	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

HO-Core-SW#

Figure 16: VLAN creation on the Head Office core switch.

10.2 Litigation VLAN Configuration

VLAN 10 was configured for the Litigation Department.

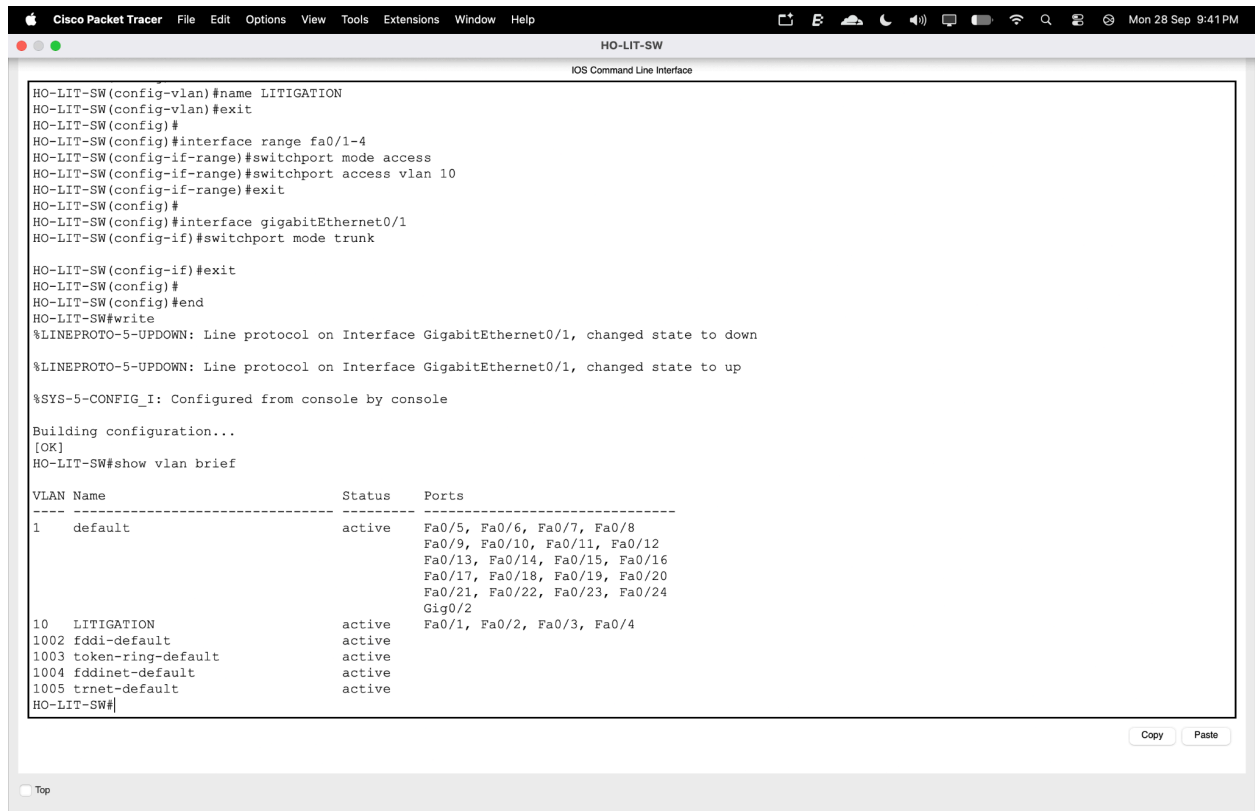
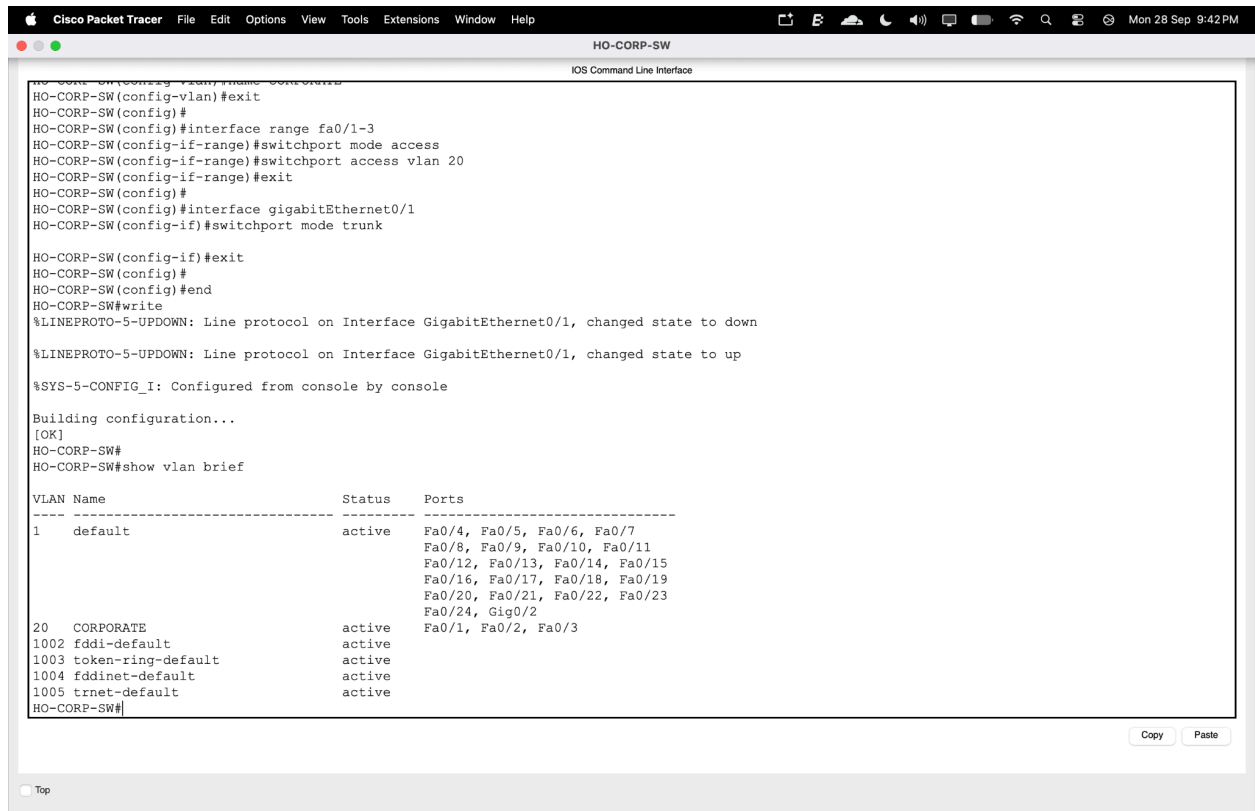


Figure 17: Litigation switch VLAN configuration.

10.3 Corporate VLAN Configuration

VLAN 20 was configured for the Corporate Law Department.



```
HO-CORP-SW(config-vlan)#exit
HO-CORP-SW(config)#
HO-CORP-SW(config)#interface range fa0/1-3
HO-CORP-SW(config-if-range)#switchport mode access
HO-CORP-SW(config-if-range)#switchport access vlan 20
HO-CORP-SW(config-if-range)#exit
HO-CORP-SW(config)#
HO-CORP-SW(config)#interface gigabitEthernet0/1
HO-CORP-SW(config-if)#switchport mode trunk

HO-CORP-SW(config-if)#exit
HO-CORP-SW(config)#
HO-CORP-SW(config)#end
HO-CORP-SW#write
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%SYS-5-CONFIG_I: Configured from console by console

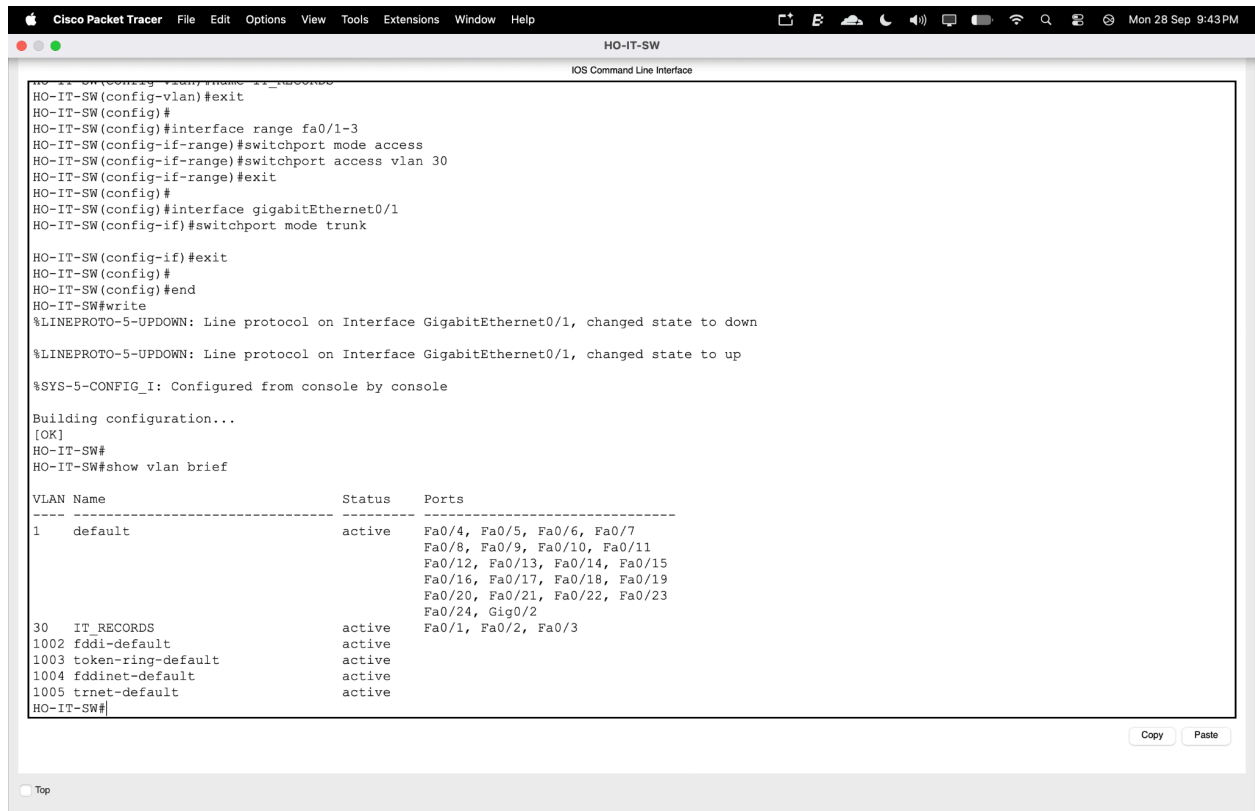
Building configuration...
[OK]
HO-CORP-SW#
HO-CORP-SW#show vlan brief

VLAN Name                Status    Ports
-----
1    default                 active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                           Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                           Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                           Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                           Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                           Fa0/24, Gig0/2
20   CORPORATE               active    Fa0/1, Fa0/2, Fa0/3
1002 fddi-default            active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default          active
HO-CORP-SW#
```

Figure 18: Corporate Law switch VLAN configuration.

10.4 IT & Records VLAN Configuration

VLAN 30 was configured for the IT & Records Department.



```
HO-IT-SW(config-vlan)#exit
HO-IT-SW(config)#
HO-IT-SW(config)#interface range fa0/1-3
HO-IT-SW(config-if-range)#switchport mode access
HO-IT-SW(config-if-range)#switchport access vlan 30
HO-IT-SW(config-if-range)#exit
HO-IT-SW(config)#
HO-IT-SW(config)#interface gigabitEthernet0/1
HO-IT-SW(config-if)#switchport mode trunk

HO-IT-SW(config-if)#exit
HO-IT-SW(config)#
HO-IT-SW(config)#end
HO-IT-SW#write
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
HO-IT-SW#
HO-IT-SW#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/2
30 IT_RECORDS	active	Fa0/1, Fa0/2, Fa0/3
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

HO-IT-SW#

Figure 19: IT & Records switch VLAN configuration.

10.5 Reception/Admin VLAN Configuration

VLAN 40 was configured for Reception/Admin devices.

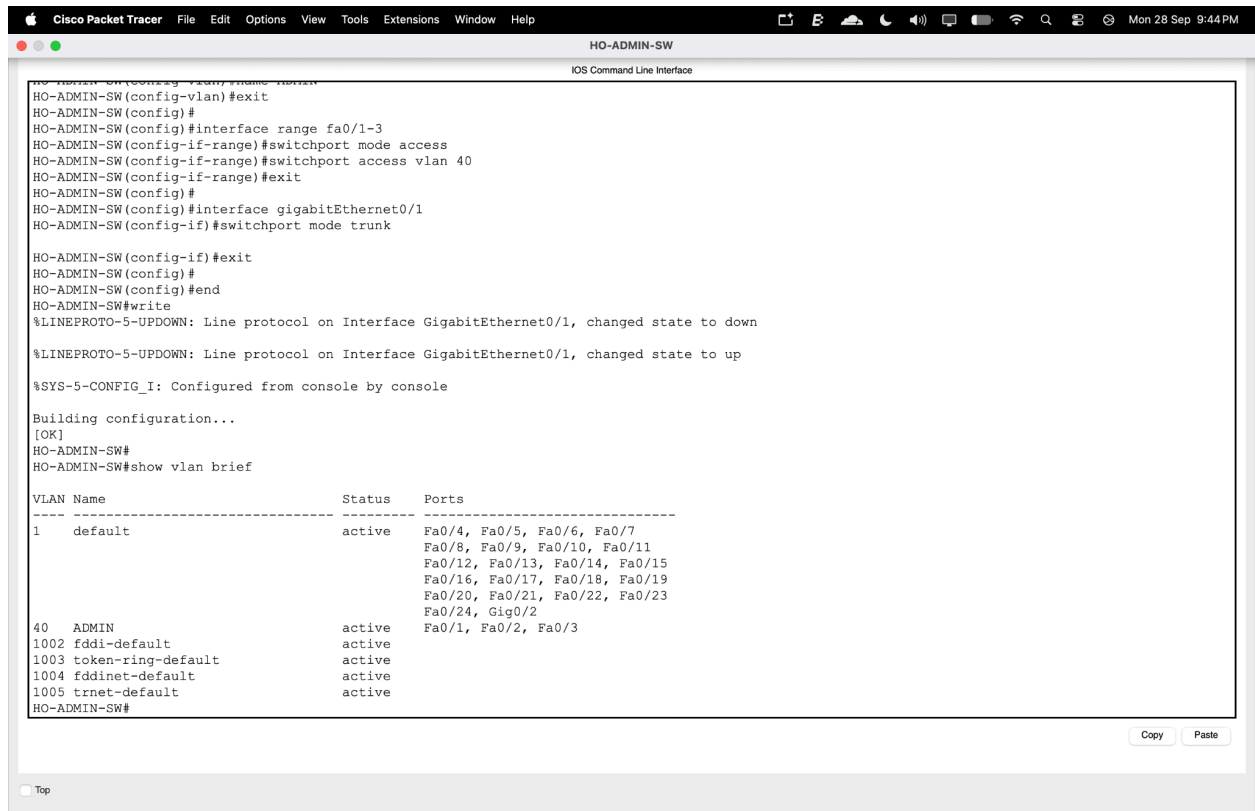


Figure 20: Reception/Admin switch VLAN configuration.

11. INTER-VLAN ROUTING IMPLEMENTATION

11.1 Router-on-a-Stick Configuration

Router-on-a-Stick was used to provide Layer 3 communication between the VLANs.

The Head Office router uses subinterfaces on **GigabitEthernet0/0**.

Subinterfac e	VLAN	Gateway	Network
G0/0.10	10	192.168.40.1	192.168.40.0/26
G0/0.20	20	192.168.40.65	192.168.40.64/26

G0/0.30	30	192.168.40.12	192.168.40.128/2
		9	8
G0/0.40	40	192.168.40.14	192.168.40.144/2
		5	8
G0/0.50	50	192.168.40.16	192.168.40.160/2
		1	8

```

HO-Router (config)#end
HO-Router#write
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/0.10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/0.20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/0.30, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/0.40, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.40, changed state to up
%SYS-5-CONFIG_I: Configured from console by console

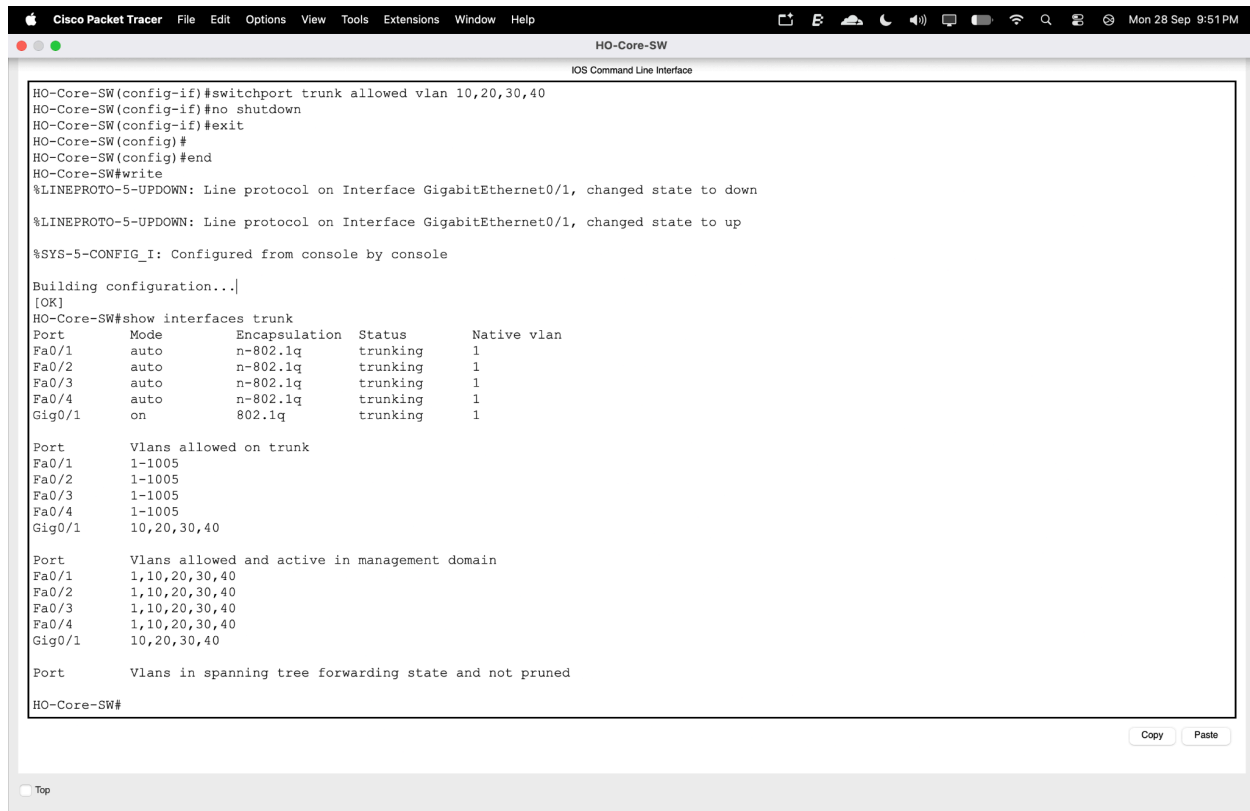
Building configuration...
[OK]
HO-Router#show ip interface brief
Interface      IP-Address      OK? Method Status  Protocol
GigabitEthernet0/0    unassigned      YES unset  up      up
GigabitEthernet0/0.10 192.168.40.1    YES manual  up      up
GigabitEthernet0/0.20 192.168.40.65   YES manual  up      up
GigabitEthernet0/0.30 192.168.40.129  YES manual  up      up
GigabitEthernet0/0.40 192.168.40.145  YES manual  up      up
GigabitEthernet0/1    unassigned      YES unset  administratively down down
GigabitEthernet0/2    unassigned      YES unset  administratively down down
Serial0/3/0          192.168.43.1    YES manual  up      up
Serial0/3/1          192.168.43.5    YES manual  up      up
Vlan1               unassigned      YES unset  administratively down down
HO-Router#

```

Figure 21: Router-on-a-Stick subinterface configuration on HO-Router.

11.2 Core Switch Trunk Configuration

The core switch trunk was configured to carry the required VLAN traffic.



```
HO-Core-SW(config-if)#switchport trunk allowed vlan 10,20,30,40
HO-Core-SW(config-if)#no shutdown
HO-Core-SW(config-if)#exit
HO-Core-SW(config)#
HO-Core-SW(config)#end
HO-Core-SW#write
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%SYS-5-CONFIG_I: Configured from console by console
Building configuration...
[OK]
HO-Core-SW#show interfaces trunk
Port      Mode      Encapsulation  Status        Native vlan
Fa0/1     auto      n-802.1q       trunking      1
Fa0/2     auto      n-802.1q       trunking      1
Fa0/3     auto      n-802.1q       trunking      1
Fa0/4     auto      n-802.1q       trunking      1
Gig0/1    on        802.1q         trunking      1

Port      Vlans allowed on trunk
Fa0/1     1-1005
Fa0/2     1-1005
Fa0/3     1-1005
Fa0/4     1-1005
Gig0/1    10,20,30,40

Port      Vlans allowed and active in management domain
Fa0/1     1,10,20,30,40
Fa0/2     1,10,20,30,40
Fa0/3     1,10,20,30,40
Fa0/4     1,10,20,30,40
Gig0/1    10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
HO-Core-SW#
```

Figure 22: Initial trunk configuration on the Head Office core switch.

12. IP ADDRESSING DESIGN

12.1 Base Network

The assigned base IPv4 network is:

192.168.40.0/22

The **/22** network provides the address space:

192.168.40.0 – 192.168.43.255

VLSM is used to divide this address space into appropriately sized subnets according to the host requirements.

13. VLSM ADDRESS ALLOCATION

13.1 LAN VLSM Address Allocation

For each LAN segment, the required host count is increased by 20% to provide spare capacity.

Network Segment	Hosts	+20%	Network / CIDR	Subnet Mask
Litigation VLAN 10	45	54	192.168.40.0/26	255.255.255.192
Corporate VLAN 20	33	40	192.168.40.64/26	255.255.255.192
IT & Records VLAN 30	9	11	192.168.40.128/28	255.255.255.240
Reception/Admin VLAN 40	4	5	192.168.40.144/28	255.255.255.240
Voice VLAN 50	VoIP	—	192.168.40.160/28	255.255.255.240

VLSM Calculation

Litigation:

$45 \times 1.20 = 54$ required hosts

A /26 provides 62 usable addresses.

Corporate Law:

$33 \times 1.20 = 39.6 \approx 40$ required hosts

A /26 provides 62 usable addresses.

IT & Records:

$9 \times 1.20 = 10.8 \approx 11$ required hosts

A /28 provides 14 usable addresses.

Reception/Admin:

$4 \times 1.20 = 4.8 \approx 5$ required hosts.

A /29 would technically provide 6 usable addresses; however, the implemented Packet Tracer network uses /28, providing 14 usable addresses and additional room for administrative devices.

Voice VLAN:

The Voice VLAN uses a /28 subnet to provide 14 usable addresses for VoIP devices.

13.2 Host Range and Gateway

Network Segment	Usable Host Range	Broadcast	Gateway
Litigation VLAN 10	192.168.40.1 – 192.168.40.62	192.168.40.63	192.168.40.1
Corporate VLAN 20	192.168.40.65 – 192.168.40.126	192.168.40.127	192.168.40.65
IT & Records VLAN 30	192.168.40.129 – 192.168.40.142	192.168.40.143	192.168.40.129
Reception/Admin VLAN 40	192.168.40.145 – 192.168.40.158	192.168.40.159	192.168.40.145
Voice VLAN 50	192.168.40.161 – 192.168.40.174	192.168.40.175	192.168.40.161

13.3 WAN Address Allocation

WAN Link	Network / CIDR	Usable Addresses	Broadcast
HO – Riverside	192.168.43.0/30	192.168.43.1 – 192.168.43.2	192.168.43.3
HO – Uptown	192.168.43.4/30	192.168.43.5 – 192.168.43.6	192.168.43.7

A /30 subnet is appropriate for a point-to-point WAN connection because it provides exactly two usable IPv4 addresses, one for each router interface.

14. END DEVICE IP ADDRESS CONFIGURATION

14.1 Litigation Device

The Litigation laptop was configured with:

Parameter	Value
Device	LIT-Laptop-01
IP Address	192.168.40.10
Subnet Mask	255.255.255.192
	2
Default Gateway	192.168.40.1

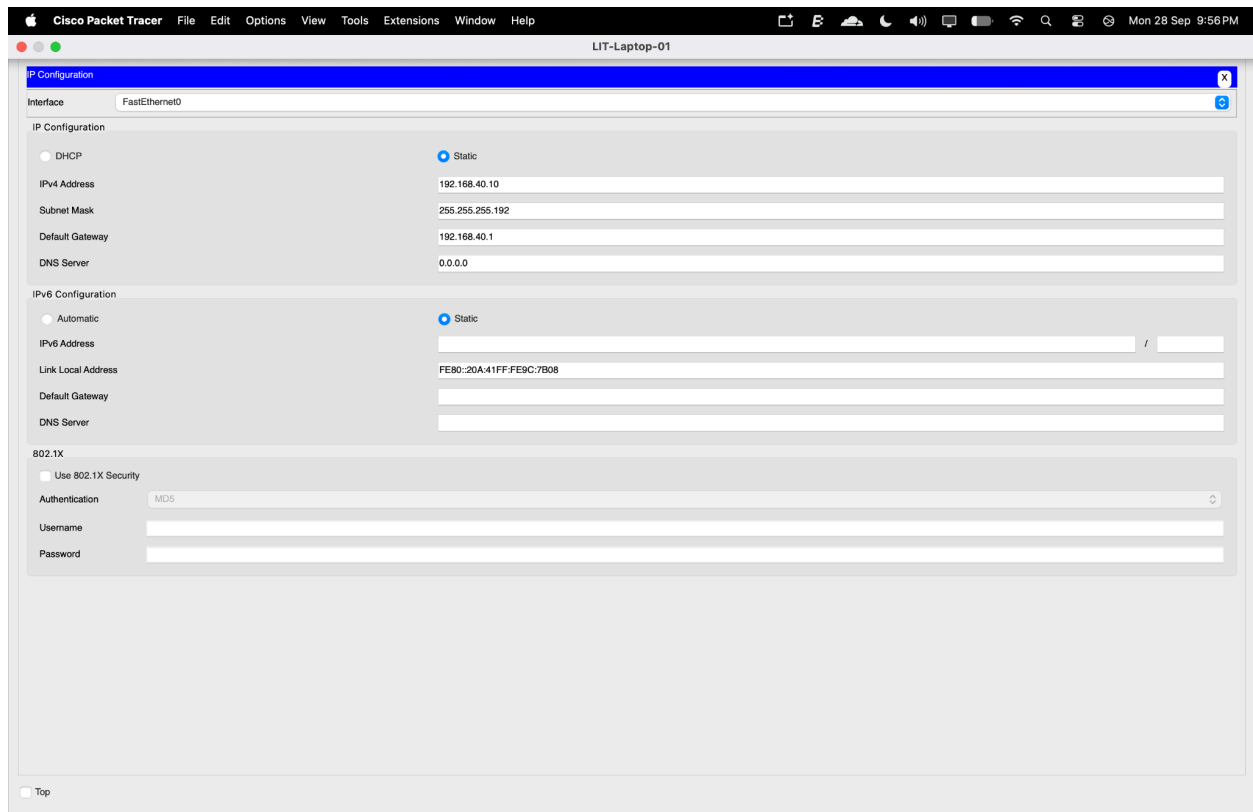


Figure 23: IP configuration of LIT-Laptop-01.

14.2 Corporate Device

The Corporate Law PC was configured with:

Parameter	Value
Device	CORP-PC-01
IP Address	192.168.40.70
Subnet Mask	255.255.255.192
Default Gateway	192.168.40.65

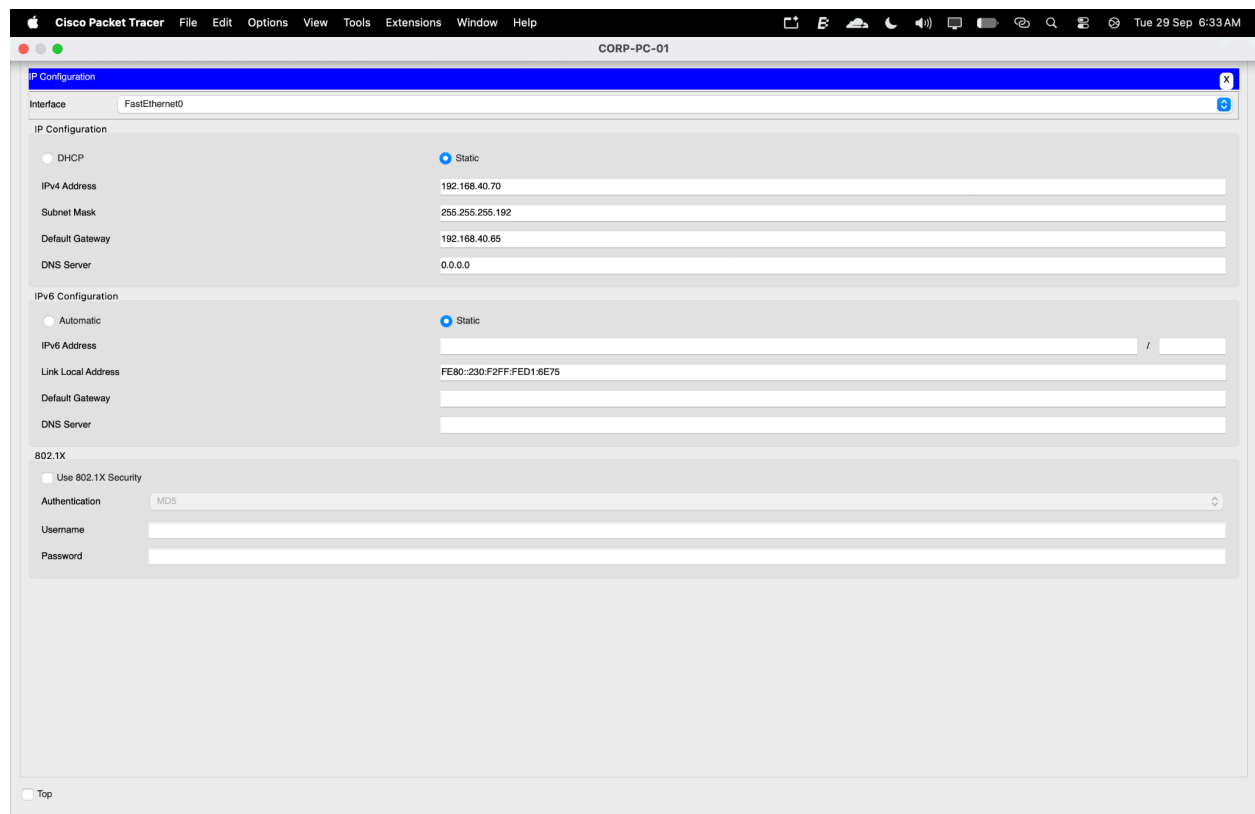


Figure 24: IP configuration of CORP-PC-01.

14.3 IT & Records Device

The IT file server was configured with:

Parameter	Value
Device	IT-File-Server
IP Address	192.168.40.130
Subnet Mask	255.255.255.240
	0
Default Gateway	192.168.40.129

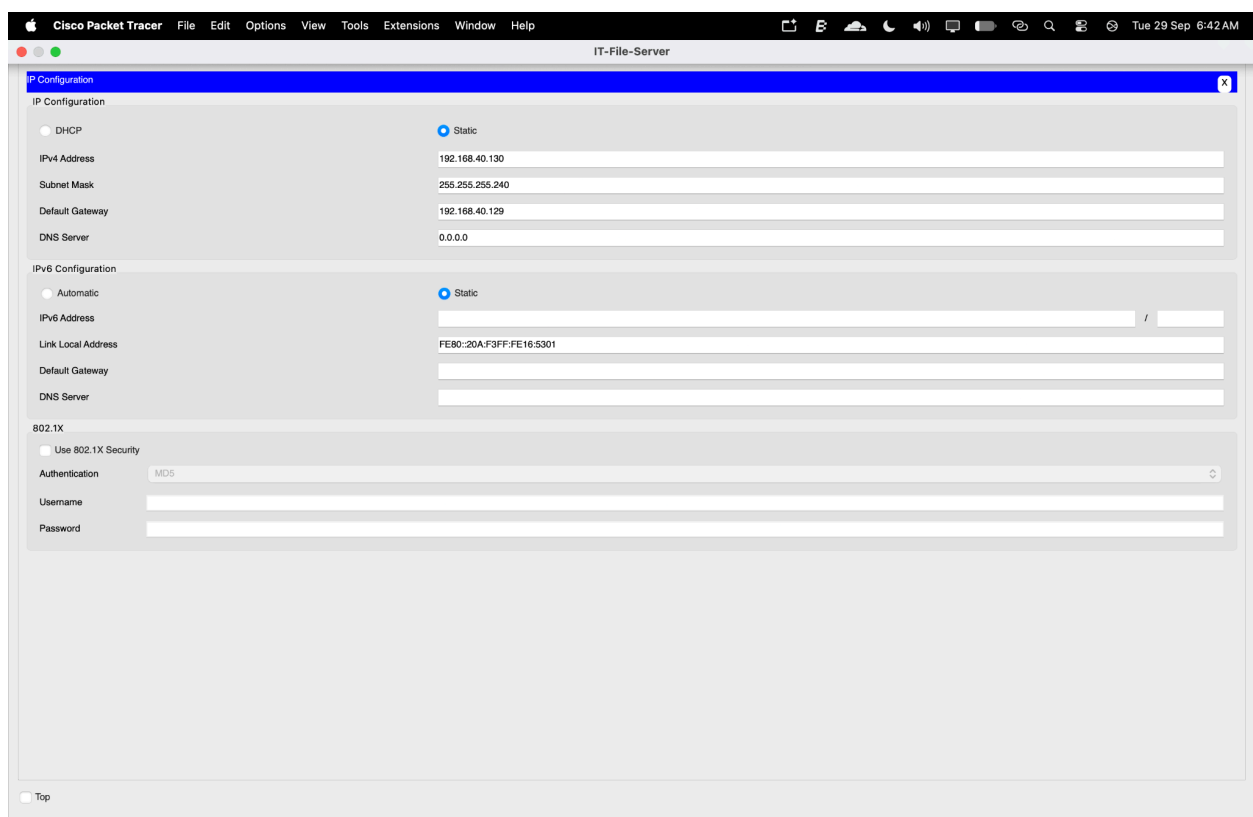


Figure 25: IP configuration of IT-File-Server.

14.4 Reception/Admin Device

The administrative PC was configured with:

Parameter	Value
-----------	-------

Device ADMIN-PC-01

IP Address 192.168.40.146

Subnet Mask 255.255.255.240

 0

Default Gateway 192.168.40.145

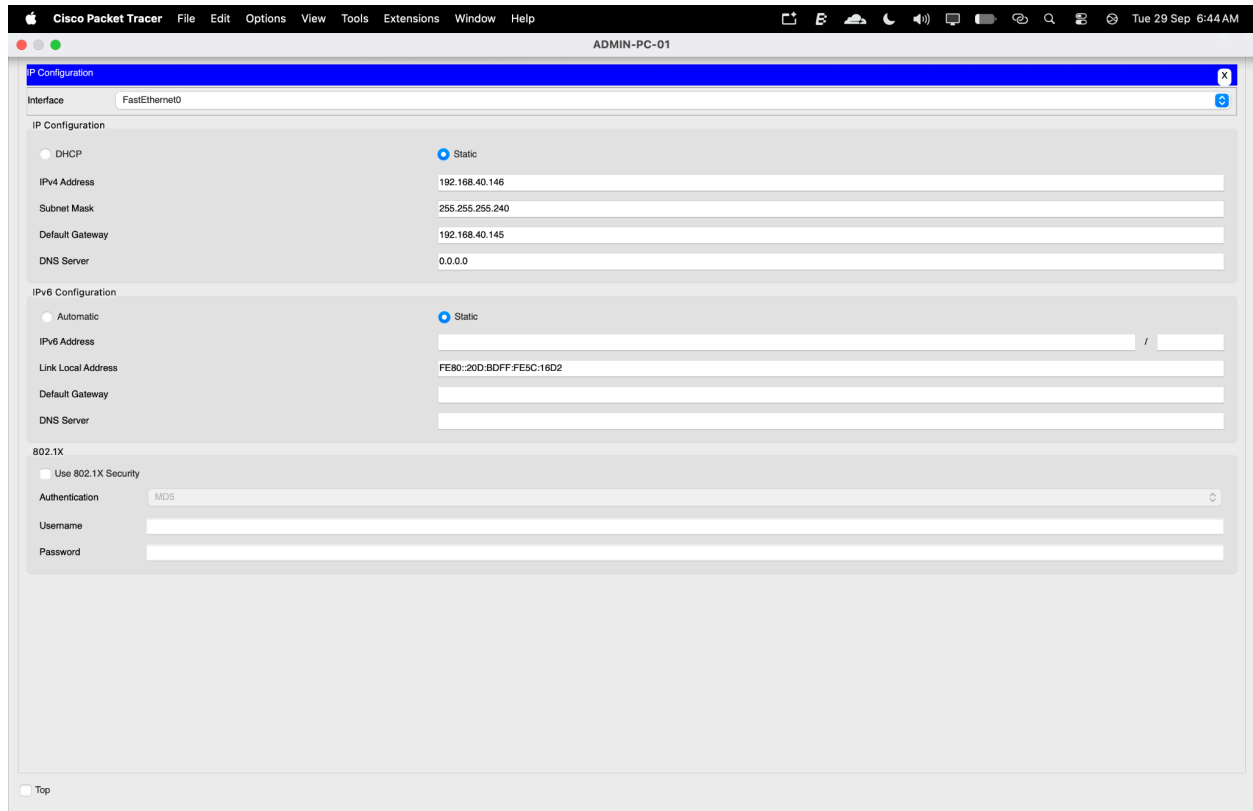


Figure 26: IP configuration of ADMIN-PC-01.

15. VOICE VLAN IMPLEMENTATION

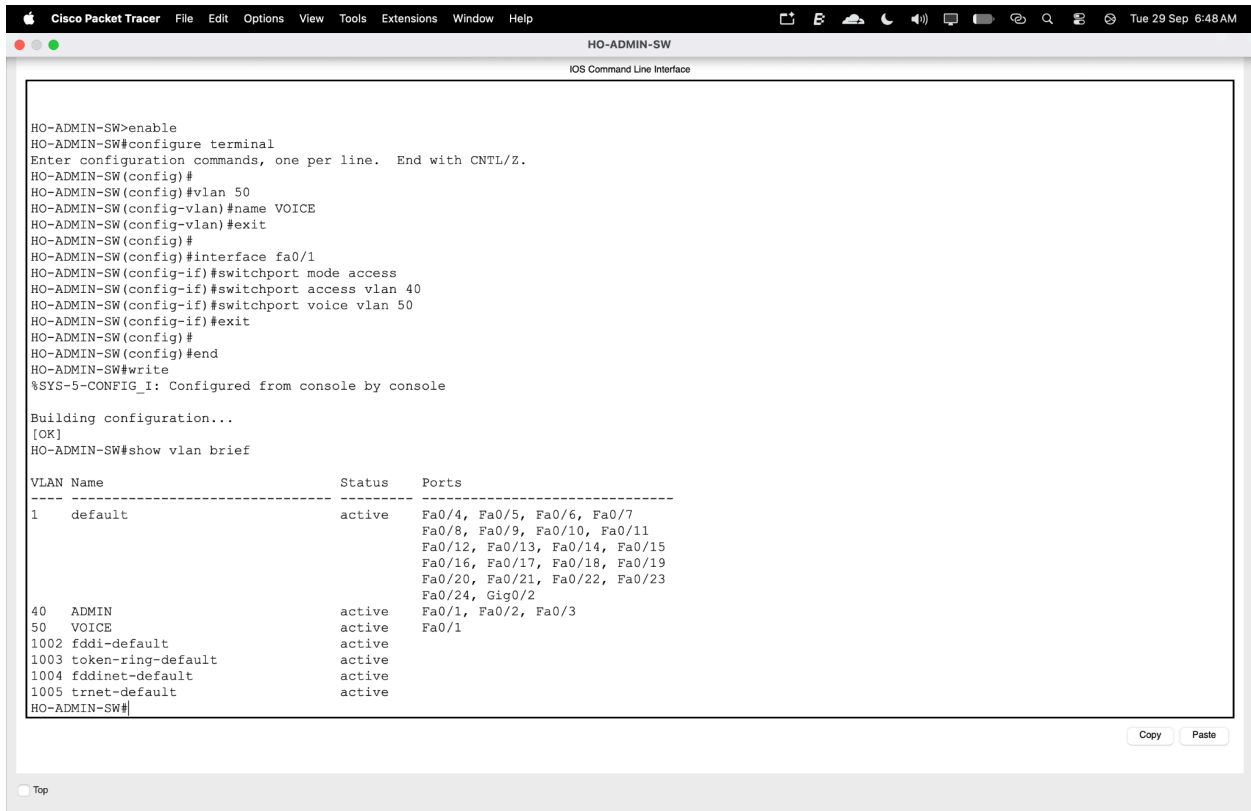
15.1 Voice VLAN Creation

A dedicated Voice VLAN was implemented using:

VLAN ID: 50

VLAN Name: VOICE

This separates VoIP traffic from normal Reception/Admin data traffic.



```
HO-ADMIN-SW>enable
HO-ADMIN-SW#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
HO-ADMIN-SW(config)#
HO-ADMIN-SW(config)#vlan 50
HO-ADMIN-SW(config-vlan)#name VOICE
HO-ADMIN-SW(config-vlan)#exit
HO-ADMIN-SW(config)#
HO-ADMIN-SW(config)#interface fa0/1
HO-ADMIN-SW(config-if)#switchport mode access
HO-ADMIN-SW(config-if)#switchport access vlan 40
HO-ADMIN-SW(config-if)#switchport voice vlan 50
HO-ADMIN-SW(config-if)#exit
HO-ADMIN-SW(config)#
HO-ADMIN-SW(config)#end
HO-ADMIN-SW#write
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
HO-ADMIN-SW#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                           Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                           Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                           Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                           Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                           Fa0/24, Gig0/2
40   ADMIN                  active    Fa0/1, Fa0/2, Fa0/3
50   VOICE                  active    Fa0/1
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default       active
1005 trnet-default         active
HO-ADMIN-SW#
```

Figure 27: Voice VLAN 50 configuration on HO-ADMIN-SW.

15.2 Voice VLAN Router Subinterface

The Head Office router was configured with:

G0/0.50

IP Address: 192.168.40.161/28

The screenshot shows the Cisco Packet Tracer interface with a terminal window for 'HO-Router'. The terminal displays the following commands and output:

```
HO-Router>enable
Password:
Password:
HO-Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
HO-Router(config)#
HO-Router(config)#interface gigabitEthernet0/0.50
HO-Router(config-subif)#encapsulation dot1Q 50
HO-Router(config-subif)#ip address 192.168.40.161 255.255.255.240
HO-Router(config-subif)#no shutdown
HO-Router(config-subif)#exit
HO-Router(config)#
HO-Router(config)#end
HO-Router#write
%LINK-5-CHANGED: Interface GigabitEthernet0/0.50, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.50, changed state to up
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
HO-Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0      unassigned      YES unset    up          up
GigabitEthernet0/0.10    192.168.40.1    YES manual   up          up
GigabitEthernet0/0.20    192.168.40.65   YES manual   up          up
GigabitEthernet0/0.30    192.168.40.129  YES manual   up          up
GigabitEthernet0/0.40    192.168.40.145  YES manual   up          up
GigabitEthernet0/0.50    192.168.40.161  YES manual   up          up
GigabitEthernet0/1      unassigned      YES unset    administratively down down
GigabitEthernet0/2      unassigned      YES unset    administratively down down
Serial0/3/0            192.168.43.1    YES manual   up          up
Serial0/3/1            192.168.43.5    YES manual   up          up
Vlan1                  unassigned      YES unset    administratively down down
HO-Router#
```

At the bottom of the terminal window, there are 'Copy' and 'Paste' buttons. Below the terminal window is a 'Top' button.

Figure 28: Router subinterface G0/0.50 for Voice VLAN 50.

15.3 Voice DHCP Pool

A DHCP pool named **VOICE_POOL** was configured for the Voice VLAN.

The available range is:

192.168.40.161 – 192.168.40.174

One address is excluded for gateway use, leaving the remaining addresses available for IP phones.

```

HO-Router#show ip dhcp pool

Pool VOICE_POOL :
  Utilization mark (high/low)      : 100 / 0
  Subnet size (first/next)         : 0 / 0
  Total addresses                   : 14
  Leased addresses                  : 0
  Excluded addresses                : 1
  Pending event                     : none

1 subnet is currently in the pool
Current index      IP address range      Leased/Excluded/Total
192.168.40.161    192.168.40.161 - 192.168.40.174  0 / 1 / 14
HO-Router#

```

Figure 29: DHCP pool configuration for Voice VLAN 50.

15.4 IP Phone DHCP Assignment

The IP phone was configured to obtain its address automatically through DHCP.

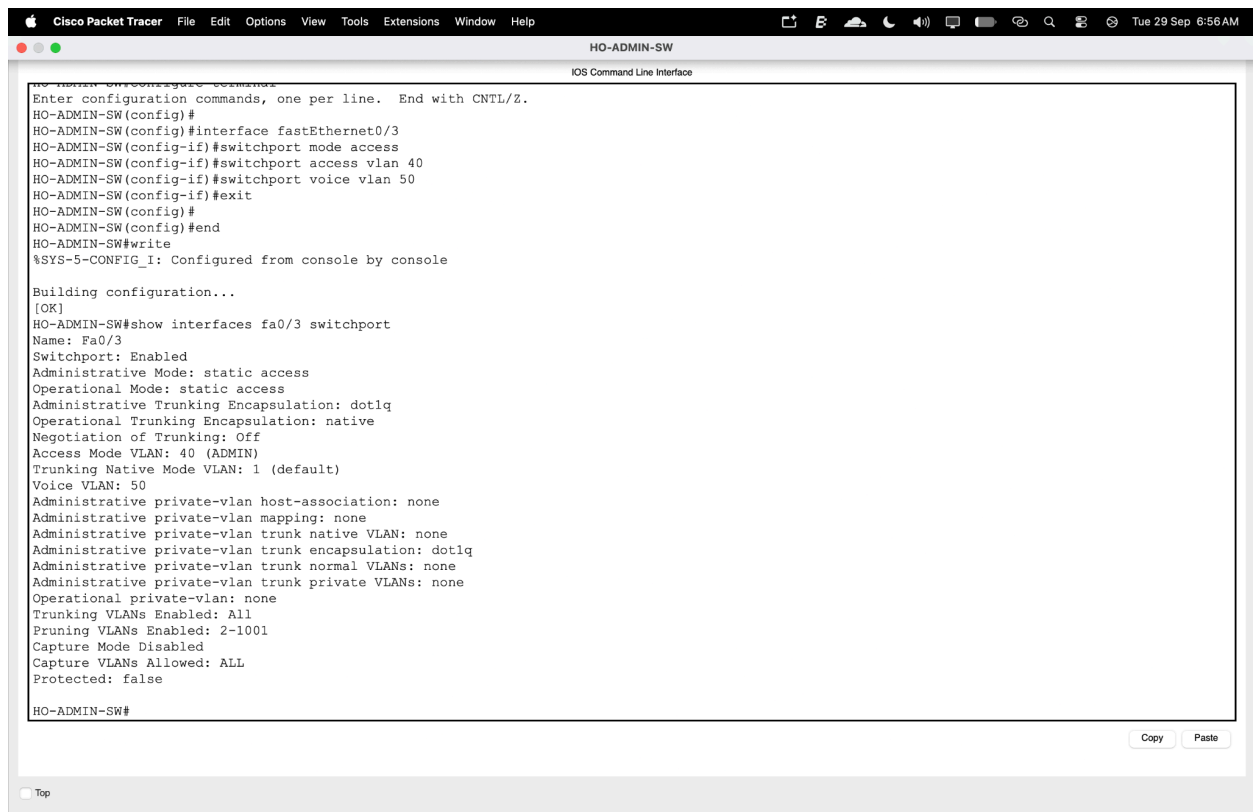


Figure 30: IP phone interface configured for DHCP.

15.5 DHCP Binding Verification

The DHCP binding table confirmed that the IP phone received:

192.168.40.162

```
HO-Router>show ip dhcp binding
IP address      Client-ID/      Lease expiration  Type
                Hardware address
192.168.40.162  00D0.5879.00C9  --               Automatic
HO-Router>
```

Figure 31: DHCP binding showing the IP address assigned to the IP phone.

16. NETWORK TESTING AND VERIFICATION

After configuration, the network was tested at multiple levels.

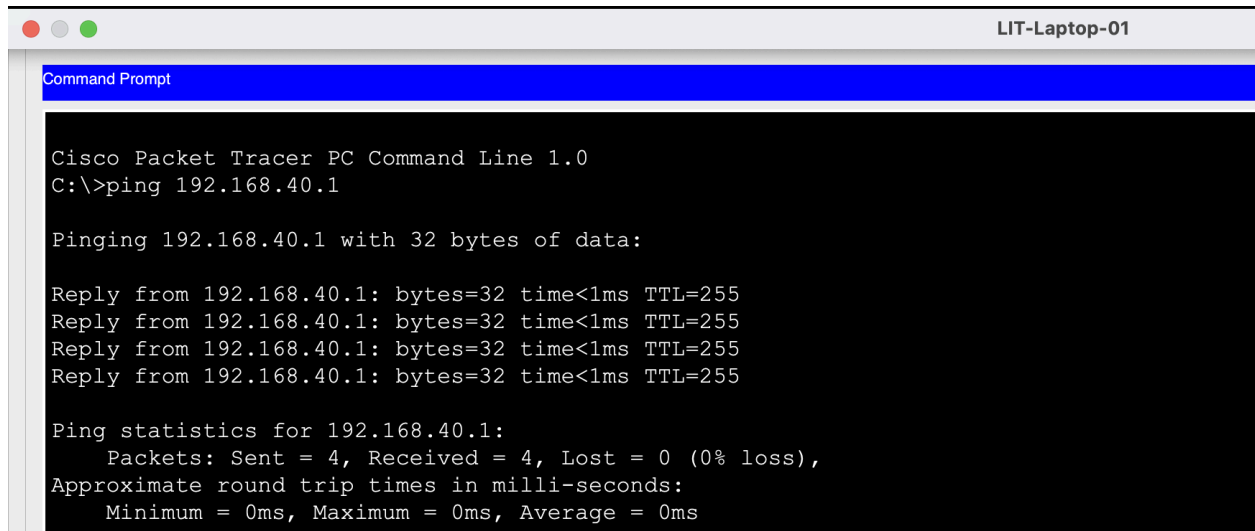
16.1 Litigation Gateway Test

The Litigation device successfully pinged its default gateway:

192.168.40.1

Result:

4/4 replies — 0% packet loss



```
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.40.1

Pinging 192.168.40.1 with 32 bytes of data:

Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.40.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 32: Successful Litigation gateway connectivity test.

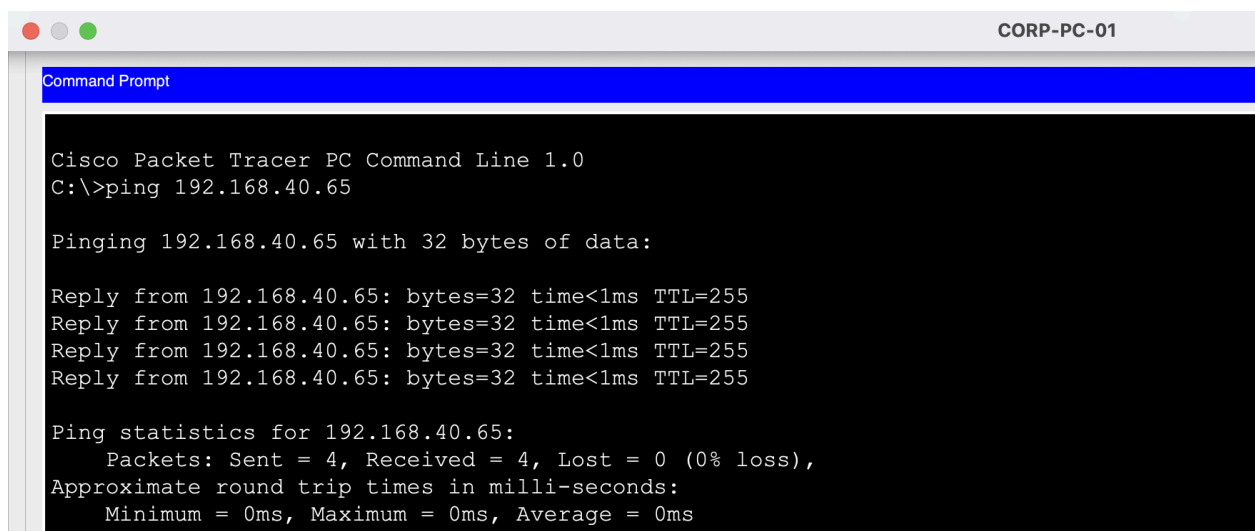
16.2 Corporate Gateway Test

The Corporate PC successfully pinged:

192.168.40.65

Result:

4/4 replies — 0% packet loss



```
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.40.65

Pinging 192.168.40.65 with 32 bytes of data:

Reply from 192.168.40.65: bytes=32 time<1ms TTL=255
Reply from 192.168.40.65: bytes=32 time<1ms TTL=255
Reply from 192.168.40.65: bytes=32 time<1ms TTL=255
Reply from 192.168.40.65: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.40.65:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 33: Successful Corporate gateway connectivity test.

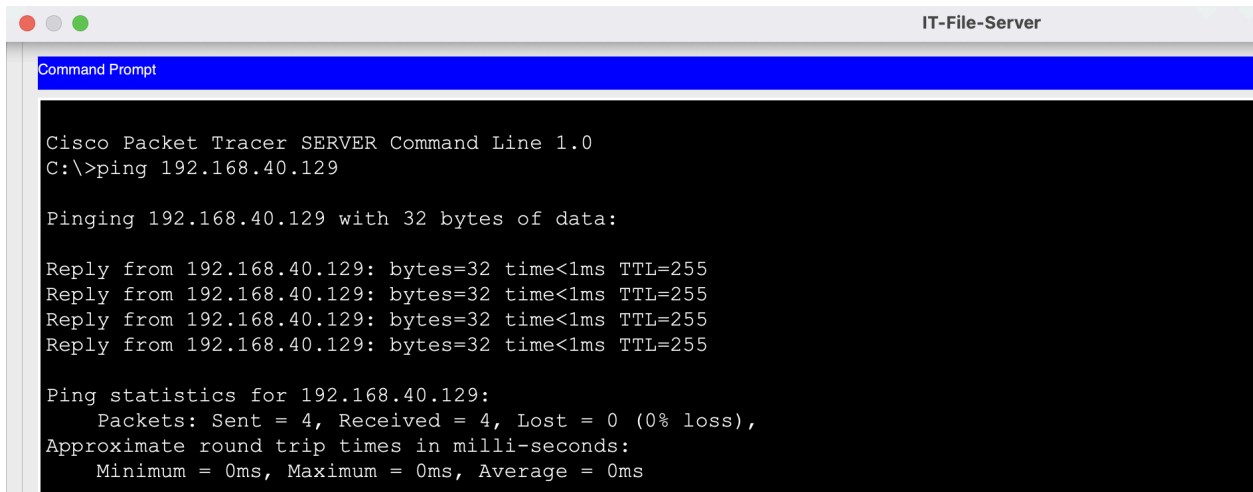
16.3 IT Gateway Test

The IT file server successfully pinged:

192.168.40.129

Result:

4/4 replies — 0% packet loss

A screenshot of a Cisco Packet Tracer window titled "IT-File-Server". Inside the window is a "Command Prompt" window with a black background and white text. The text shows the execution of a ping command to 192.168.40.129, resulting in four successful replies with 0% packet loss.

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 192.168.40.129

Pinging 192.168.40.129 with 32 bytes of data:

Reply from 192.168.40.129: bytes=32 time<1ms TTL=255
Reply from 192.168.40.129: bytes=32 time<1ms TTL=255
Reply from 192.168.40.129: bytes=32 time<1ms TTL=255
Reply from 192.168.40.129: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.40.129:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 34: Successful IT & Records gateway connectivity test.

16.4 Admin Gateway Test

The Admin PC successfully pinged:

192.168.40.145

Result:

4/4 replies — 0% packet loss

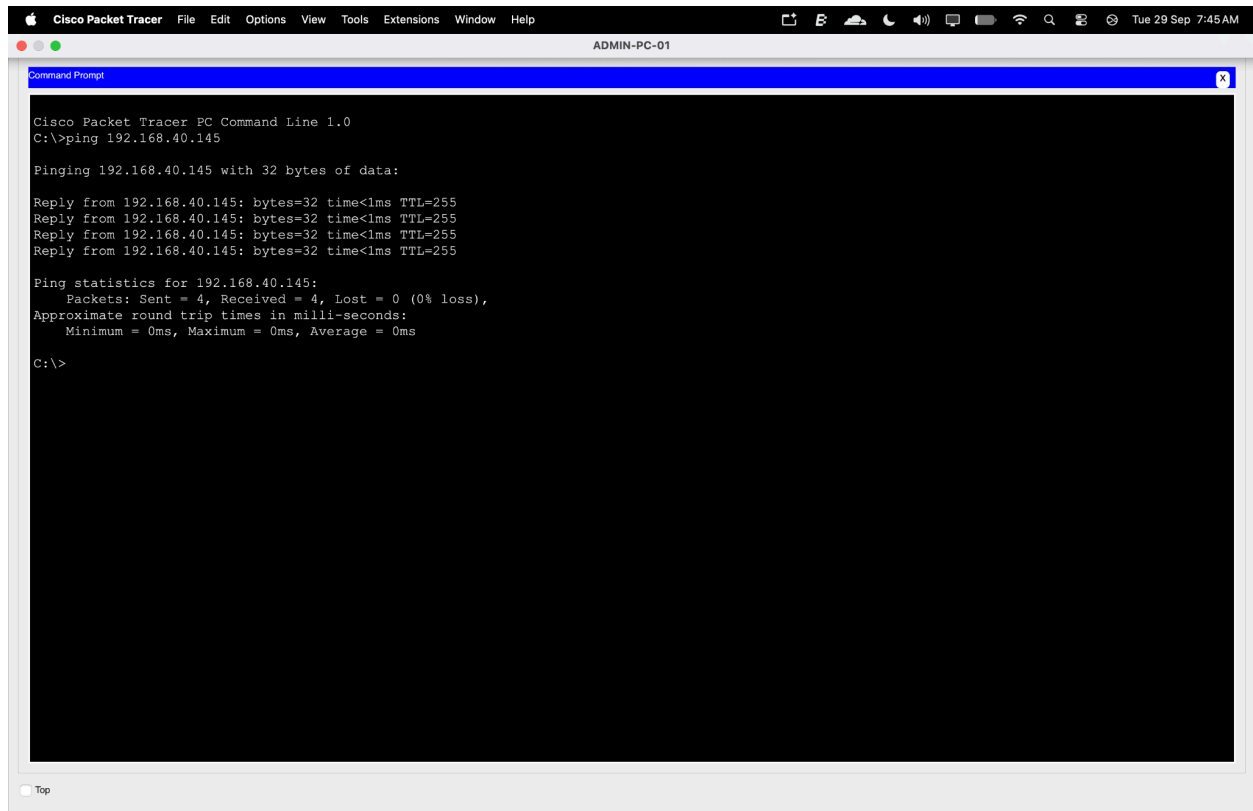


Figure 35: Successful Reception/Admin gateway connectivity test.

16.5 Inter-VLAN Test: Litigation to IT

The Litigation device successfully communicated with the IT file server:

Destination: 192.168.40.130

Result:

4/4 replies — 0% packet loss

This confirms that Router-on-a-Stick is providing communication between the Litigation and IT VLANs.

```
C:\>ping 192.168.40.130

Pinging 192.168.40.130 with 32 bytes of data:

Reply from 192.168.40.130: bytes=32 time<1ms TTL=127
Reply from 192.168.40.130: bytes=32 time<1ms TTL=127
Reply from 192.168.40.130: bytes=32 time<1ms TTL=127
Reply from 192.168.40.130: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.40.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 36: Successful inter-VLAN connectivity from Litigation to IT & Records.

16.6 Inter-VLAN Test: Admin to Corporate

The Admin device successfully communicated with the Corporate Law device:

Destination: 192.168.40.70

Result:

4/4 replies — 0% packet loss

```
C:\>ping 192.168.40.70

Pinging 192.168.40.70 with 32 bytes of data:

Reply from 192.168.40.70: bytes=32 time<1ms TTL=127
Reply from 192.168.40.70: bytes=32 time=1ms TTL=127
Reply from 192.168.40.70: bytes=32 time<1ms TTL=127
Reply from 192.168.40.70: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.40.70:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Figure 37: Successful inter-VLAN connectivity from Admin to Corporate Law.

16.7 WAN Test: Head Office to Riverside

The Head Office router successfully pinged:

192.168.43.2

Result:

100% successful

```
HO-Router>ping 192.168.43.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.43.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/4/15 ms
```

Figure 38: Successful WAN connectivity from HO-Router to Riverside-Router.

16.8 WAN Test: Head Office to Uptown

The Head Office router successfully pinged:

192.168.43.6

Result:

100% successful

```
HO-Router>ping 192.168.43.6

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.43.6, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 3/5/13 ms
```

Figure 39: Successful WAN connectivity from HO-Router to Uptown-Router.

16.9 Routing Table Verification

The Head Office router routing table confirmed the directly connected VLAN networks and WAN networks.

Important connected networks include:

- 192.168.40.0/26
- 192.168.40.64/26
- 192.168.40.128/28
- 192.168.40.144/28
- 192.168.40.160/28
- WAN subnets under 192.168.43.0

```
HO-Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```
    192.168.40.0/24 is variably subnetted, 10 subnets, 3 masks
C       192.168.40.0/26 is directly connected, GigabitEthernet0/0.10
L       192.168.40.1/32 is directly connected, GigabitEthernet0/0.10
C       192.168.40.64/26 is directly connected, GigabitEthernet0/0.20
L       192.168.40.65/32 is directly connected, GigabitEthernet0/0.20
C       192.168.40.128/28 is directly connected, GigabitEthernet0/0.30
L       192.168.40.129/32 is directly connected, GigabitEthernet0/0.30
C       192.168.40.144/28 is directly connected, GigabitEthernet0/0.40
L       192.168.40.145/32 is directly connected, GigabitEthernet0/0.40
C       192.168.40.160/28 is directly connected, GigabitEthernet0/0.50
L       192.168.40.161/32 is directly connected, GigabitEthernet0/0.50
    192.168.43.0/24 is variably subnetted, 4 subnets, 2 masks
```

Figure 40: HO-Router routing table showing connected VLAN and WAN networks.

17. NETWORK PROTOCOL EXPLANATION

17.1 DHCP Process

Dynamic Host Configuration Protocol (DHCP) automatically provides network configuration information to devices.

The basic process is:

1. **DHCP Discover** — The client broadcasts a request for an IP configuration.
2. **DHCP Offer** — The DHCP server offers an available IP address.
3. **DHCP Request** — The client requests the offered configuration.
4. **DHCP Acknowledgement** — The DHCP server confirms the assignment.

In this project, DHCP was used for automatic IP assignment of the VoIP phone.

17.2 ARP Process

Address Resolution Protocol (ARP) maps an IPv4 address to the corresponding MAC address on a local network.

When a device needs to communicate:

1. It checks its ARP cache.
2. If the destination MAC address is unknown, an ARP request is broadcast.
3. The device owning the requested IP address responds with its MAC address.
4. The sender can then construct and transmit the Ethernet frame.

If the destination is on another network, the sender normally resolves the MAC address of its default gateway.

17.3 DNS Process

Domain Name System (DNS) translates human-readable hostnames into IP addresses.

For example:

Case-management portal hostname

↓

IP Address

DNS allows users to access services using names rather than remembering numerical IP addresses.

17.4 ICMP Process

Internet Control Message Protocol (ICMP) is used for network diagnostic and error-reporting functions.

The **ping** command uses ICMP Echo Request and Echo Reply messages.

A successful ping can verify:

- Reachability
- IP connectivity
- Basic routing
- Gateway availability

18. DHCP → ARP → DNS → ICMP SEQUENCE

When a new workstation is connected to the enterprise network and needs to access an internal case-management portal, several network processes may occur.

First, the workstation obtains its network configuration using **DHCP**. It broadcasts a DHCP Discover message, receives a DHCP Offer, sends a DHCP Request, and receives a DHCP Acknowledgement containing an IP address, subnet mask, default gateway, and other configuration information.

Next, the workstation may use **ARP** to determine the MAC address required for local communication. If the destination is outside the workstation's subnet, the workstation resolves the MAC address of its default gateway.

When the user enters the internal case-management portal hostname, **DNS** can translate that hostname into the corresponding server IP address.

After the destination IP address is known, normal application communication can begin. **ICMP** is not required for ordinary web browsing, but it is commonly used to verify connectivity using the **ping** command. A successful ICMP Echo Reply confirms basic IP reachability between the source and destination.

Therefore, DHCP provides configuration, ARP resolves local Layer 2 addressing, DNS resolves hostnames, and ICMP provides a useful diagnostic mechanism.

19. OSI / TCP-IP MODEL MAPPING

The following table maps common day-to-day network activities to the OSI and TCP/IP models.

Day-to-Day Activity	OSI Layer	TCP/IP Layer	Explanation
Lawyer accesses internal case-management portal	Layer 7 – Application	Application	Application protocols provide access to the portal service.
Switch forwards an Ethernet frame using MAC address	Layer 2 – Data Link	Network Access	The switch uses MAC addresses to forward frames.
Router forwards traffic to another office	Layer 3 – Network	Internet	IP addressing allows the router to select the destination network.
New workstation receives network configuration	Layer 7 – Application	Application	DHCP provides IP configuration to the client.
Administrator performs a ping test	Layer 3 – Network	Internet	ICMP is used to test IP reachability.

20. OSI LAYER DEVICE MAPPING

Device	Primary OSI Layer	Role in Project
End Devices	Layers 1–7	Generate and receive application and network traffic
Access Switches	Layer 2	VLAN and MAC-based frame forwarding
Core Switch	Layer 2	Central switching and VLAN trunk connectivity
Router	Layer 3	Inter-VLAN routing and WAN connectivity

Serial WAN
Interface

Layers 1–2

Physical and data-link communication over serial
connection

The switches primarily operate at Layer 2 in this implementation, while the Head Office router performs the Layer 3 routing function.

21. PRIVATE AND PUBLIC IP ADDRESSING

The implemented internal network uses **private IPv4 addressing**.

The primary internal address space is:

192.168.40.0/22

The WAN point-to-point links also use private addresses:

- 192.168.43.0/30
- 192.168.43.4/30

These addresses belong to the private IPv4 address range 192.168.0.0/16.

Private addresses are appropriate for internal organizational networks because they are not directly routable over the public Internet.

Advantages of Private Addressing

- Conserves globally routable public IPv4 addresses
- Provides an organized internal addressing structure
- Supports internal network segmentation
- Allows address reuse within private networks
- Works well with NAT when Internet access is required

Private WAN Addressing

Using private IP addresses on the point-to-point WAN links is appropriate because the leased connections are used for communication between the organization's routers.

However, private addressing itself should not be considered encryption. Security controls such as ACLs, firewalls, authentication, or encryption may be required depending on the organization's security requirements.

If Internet connectivity is introduced in the future, public addressing can be used at the appropriate edge interface, generally together with NAT/PAT.

22. COMPLETE IP ADDRESSING SUMMARY

22.1 Head Office VLAN Gateways

VLAN	Name	Network	Gateway	Mask
10	LITIGATION	192.168.40.0/26	192.168.40.1	255.255.255.192
20	CORPORATE	192.168.40.64/26	192.168.40.65	255.255.255.192
30	IT	192.168.40.128/28	192.168.40.129	255.255.255.240
40	ADMIN	192.168.40.144/28	192.168.40.145	255.255.255.240
50	VOICE	192.168.40.160/28	192.168.40.161	255.255.255.240

22.2 Router Interfaces

Device	Interface	IP Address
HO-Router	G0/0.10	192.168.40.1/26
HO-Router	G0/0.20	192.168.40.65/26
HO-Router	G0/0.30	192.168.40.129/28

HO-Router	G0/0.40	192.168.40.145/28
HO-Router	G0/0.50	192.168.40.161/28
HO-Router	S0/3/0	192.168.43.1/30
HO-Router	S0/3/1	192.168.43.5/30
Riverside-Router	S0/1/0	192.168.43.2/30
Uptown-Router	S0/1/0	192.168.43.6/30

22.3 Tested End Devices

Device	IP Address	Mask	Gateway
LIT-Laptop-01	192.168.40.10	/26	192.168.40.1
CORP-PC-01	192.168.40.70	/26	192.168.40.65
IT-File-Server	192.168.40.130	/28	192.168.40.129
ADMIN-PC-01	192.168.40.146	/28	192.168.40.145
IP Phone	192.168.40.162	/28	192.168.40.161

23. FINAL NETWORK TOPOLOGY

After completing the configurations and verification tests, the complete enterprise network topology was captured.

The final implementation contains:

- Head Office
- Litigation Department
- Corporate Law Department
- IT & Records Department
- Reception/Admin Department

- Voice VLAN
- Head Office router
- Core switch
- Departmental access switches
- Riverside branch
- Uptown branch
- Serial WAN connections
- End-user devices
- Printers
- IP phone
- VLAN segmentation
- Inter-VLAN routing
- WAN connectivity

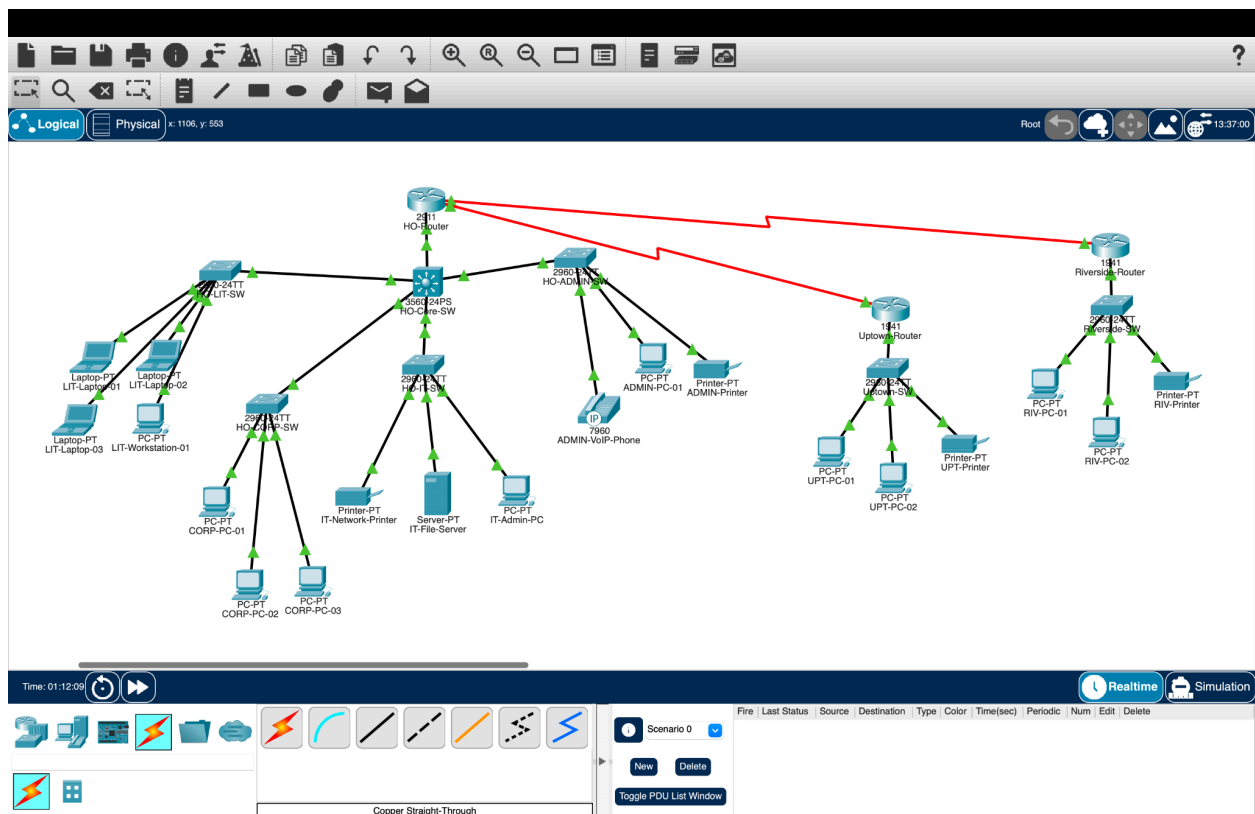


Figure 41: Final completed Harbor Lane Legal Group network topology showing the Head Office, Riverside branch, Uptown branch, VLAN-based departmental segmentation, WAN links, and connected end devices.

This screenshot should preferably be placed on its **own page** because it is the final logical network diagram and is one of the most important figures in the report.

24. CONCLUSION

The enterprise network for Harbor Lane Legal Group was successfully designed and implemented using Cisco Packet Tracer.

The project implemented departmental VLAN segmentation, VLSM-based IP addressing, Router-on-a-Stick inter-VLAN routing, trunk connectivity, Voice VLAN functionality, DHCP-based IP assignment for the IP phone, and serial WAN connectivity between the Head Office and branch offices.

The addressing plan was designed using the 192.168.40.0/22 base network and divided using VLSM according to the host requirements. Appropriate subnet sizes were selected for the Litigation, Corporate Law, IT & Records, Reception/Admin, Voice, and point-to-point WAN networks.

The implementation was verified using gateway ping tests, inter-VLAN connectivity tests, WAN ping tests, DHCP binding verification, and routing-table inspection. The successful test results demonstrate that the configured network components are communicating according to the implemented design.

The final topology provides a structured and scalable foundation for the organization's network requirements.

25. FUTURE IMPROVEMENTS

The implemented network can be further enhanced with additional enterprise networking and security technologies.

Possible future improvements include:

- Firewall implementation
- Access Control Lists (ACLs)
- Wireless network infrastructure
- Network monitoring
- Redundant network links
- Dynamic routing protocols such as OSPF
- Centralized authentication
- Network logging and monitoring

- Backup connectivity
- Additional security policies

These improvements can further strengthen network security, reliability, manageability, and scalability.

END OF REPORT
